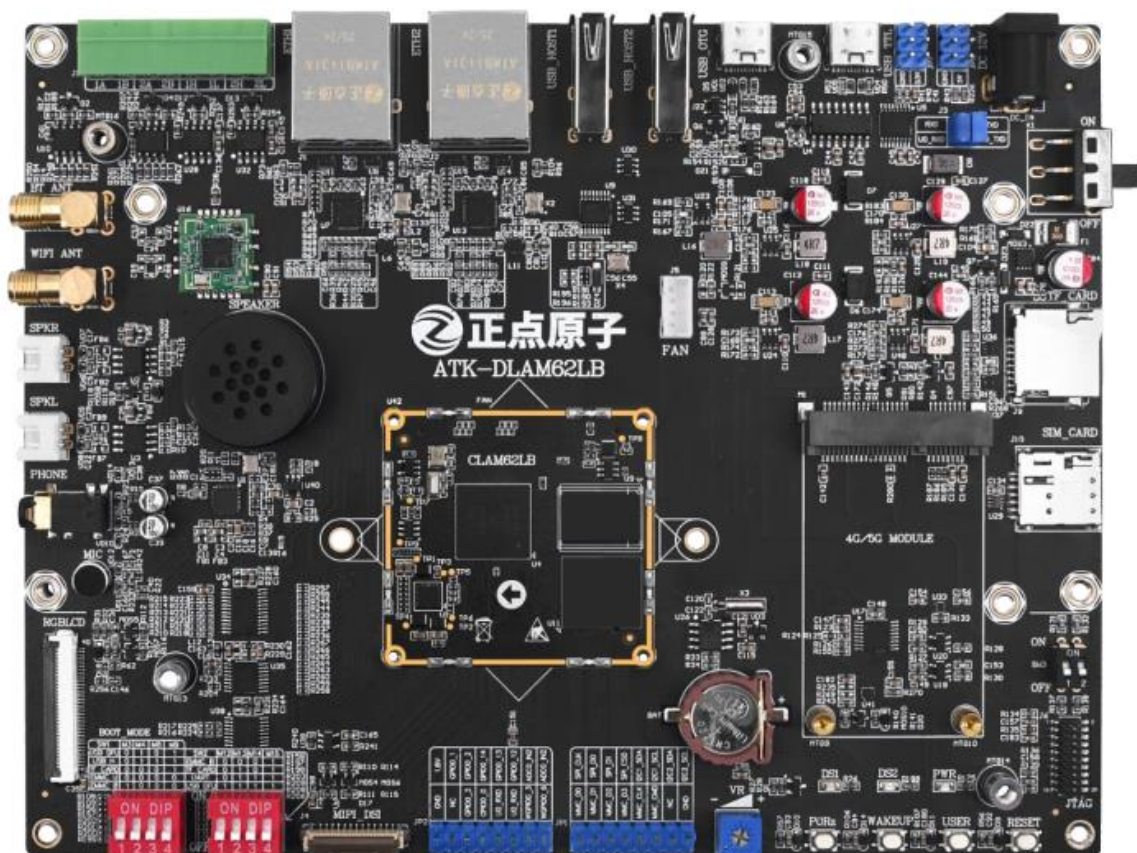


ATK-DLAM62L

Quick test manual

V1.0



1. Shopping:TMALL: <https://zhengdianyuanzi.tmall.com>TAOBAO: <https://openedv.taobao.com>**2. Download**Address: <http://www.openedv.com/docs/index.html>**3. FAE**Website : www.alientek.comForum : <http://www.openedv.com/forum.php>Videos : www.yuanzige.com

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In order to get the latest version of product information, please regularly visit the download center or contact the customer service of Taobao ALIENTEK flagship store. Thank you for your tolerance and support.

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Brief

This document is intended for users to experience the ALIENTEK AM62L development board and can be regarded as a hardware test document.

It allows users to learn how to use the development board, common test methods, as well as the on-board resources, making it a relatively important document.

This document is similar to other ALIENTEK quick start documents, and it is believed that users already have some understanding of Linux testing methods.

If there are any errors or omissions, please correct them on the ALIENTEK forum or contact the author of this document via QQ: 820562576.

Environment used in this document:

- Windows 10 64-bit; also compatible with Windows 7 to Windows 10.

(Windows 11 has not been tested, as all development is done under Windows 10;

however, it should theoretically work with Windows 11.) This document is written based on a 64-bit operating system.

- Ubuntu 22.04. Ubuntu must be version 22.04.
- Newer Ubuntu versions have different installation environments that may cause errors,
- which you will need to resolve on your own!
- Readers are expected to know how to transfer files between Ubuntu and Windows
- using FileZilla, WinSCP, and Git for Windows.
- This document is written specifically for the ALIENTEK ATK-DLAM62L Buildroot system.

If you use a different file system, experiments may fail due to missing commands or modules.

Please take note!

Chapter 1. Introduces ATK-DLAM62L hardware

resources

The development board adopts a design combining a baseboard and a core board, connected in a board-to-board manner. The core board features a small footprint and excellent stability. The development board is powered by a 12V/1A power adapter.

In terms of physical interfaces, the development board is equipped with two display interfaces: RGB and MIPI DSI respectively. It also provides two audio input interfaces and two audio output interfaces.

In addition, it integrates one Gigabit Ethernet port, two RS-485 interfaces, and two CAN interfaces, making it highly suitable for industrial applications.

The baseboard resource diagram of the ATK-DLAM62L development board is shown as follows:

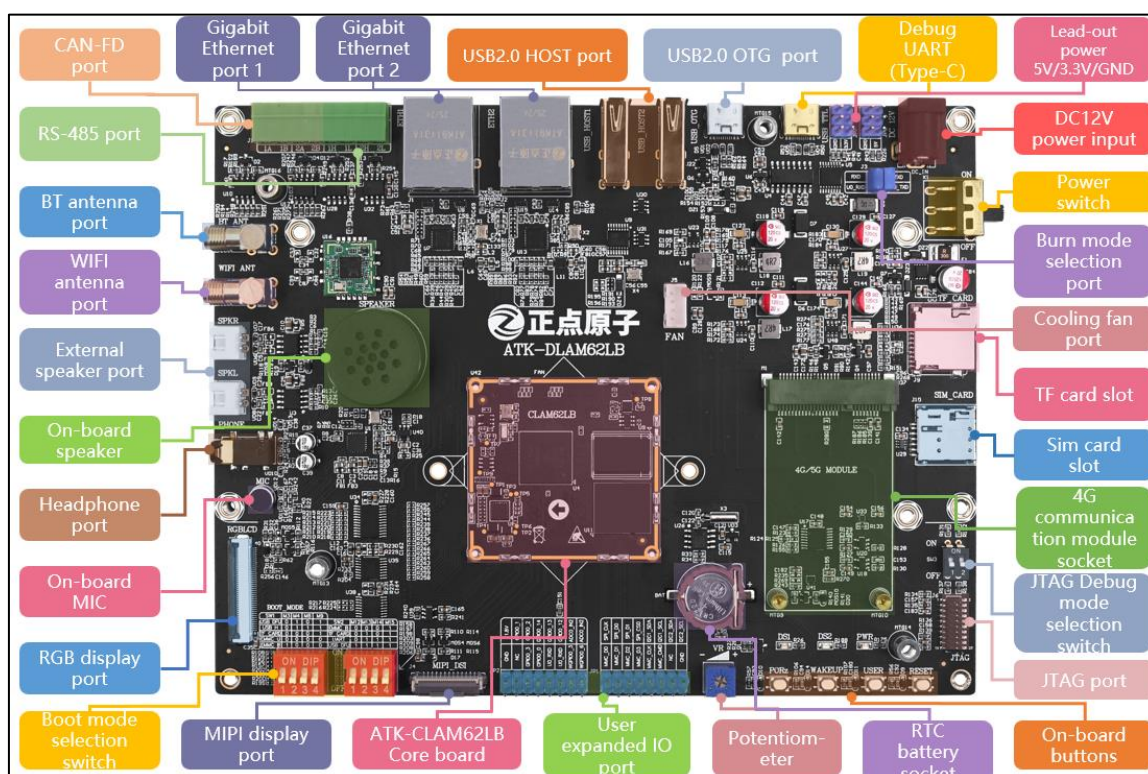


Figure 2.1-1 ATK-DLAM62L development board resource diagram

Chapter 2. Preparation of ATK-DLAM62L development board before use

This chapter is briefly introduced as follows:

- Install the serial port debugging driver for the development board and learn how to use the serial port debugging host computer.
- Log in to the development board.

2.1 Boot Mode Description

The circuit of the boot mode configuration port on the ATK-DLAM62L development board is shown in the figure.

It should be noted that the pull-up level of these pins is provided by **VCC_3V3_MAIN**.

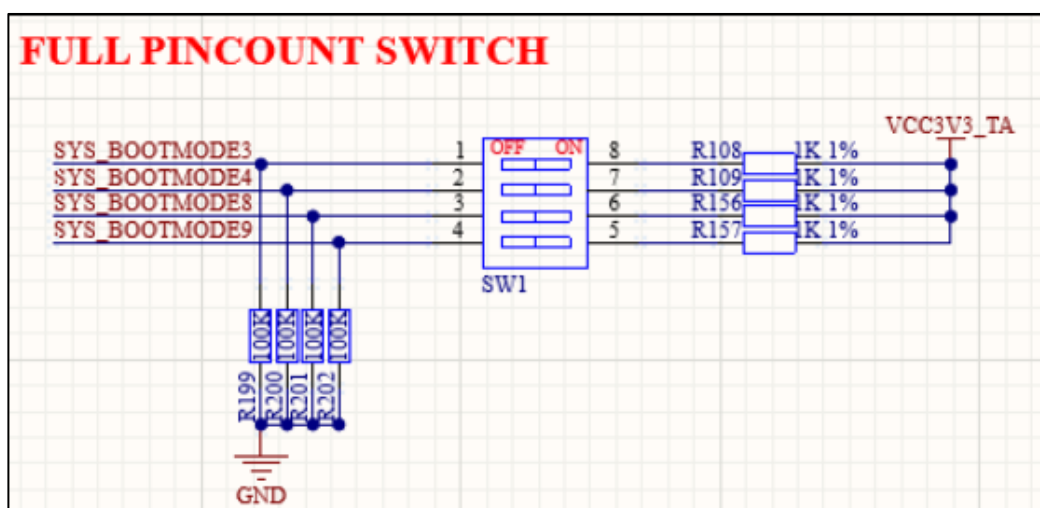


Figure 2.1-1 FULL PINCOUNT mode

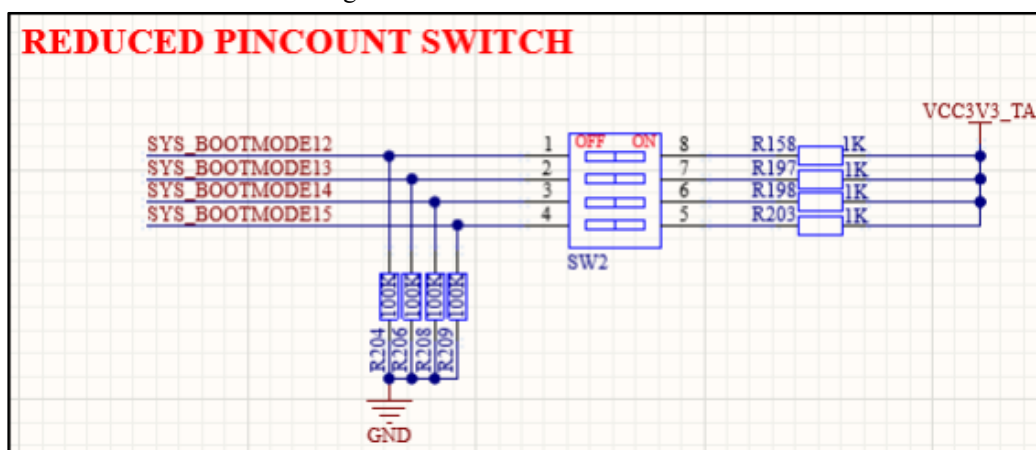


Figure 2.1-2 REDUCED PINCOUNT mode

The ATK-DLAM62L supports two distinct boot pin selection modes: FULL PINCOUNT mode and REDUCED PINCOUNT mode.

In FULL PINCOUNT mode, all boot pins **BOOTMODE[0:15]** require pull-up or pull-down configuration.

In REDUCED PINCOUNT mode, only 4 boot pins **BOOTMODE[12:15]** need to be configured. These two modes do not interfere with each other.

For both modes, it is necessary to ensure that the levels of the corresponding BOOTMODE pins are not disturbed by external devices during the CPU power-on initialization process.

Otherwise, the CPU may fail to boot normally.

The AM62L supports booting from a variety of different devices. The boot device can be selected by setting the DIP switches.

In FULL PINCOUNT mode, it corresponds to the SW1 DIP switch on the baseboard.

(The remaining pins on the baseboard are connected to pull-up and pull-down resistors.)

When using this mode, set all the SW2 DIP switches to 0.

Boot Mode(SW1)	M3	M4	M8	M9
USB(DFU)	0	1	0	1
USB(HOST)	0	1	1	1
MicroSD	0	0	0	1
EMMC(UDA)	0	0	0	0
EMMC(BOOT)	1	0	0	0

REDUCED PINCOUNT mode corresponds to the SW2 DIP switch on the baseboard.

Boot Mode(SW2)	M12	M13	M14	M15
USB(HOST)	0	0	0	1
MicroSD	0	1	0	1
UART	0	1	1	1
USB0 DFU	1	1	1	1

The ALIENTEK ATK-DLAM62L development board supports three boot modes: USB, SD card, and eMMC.

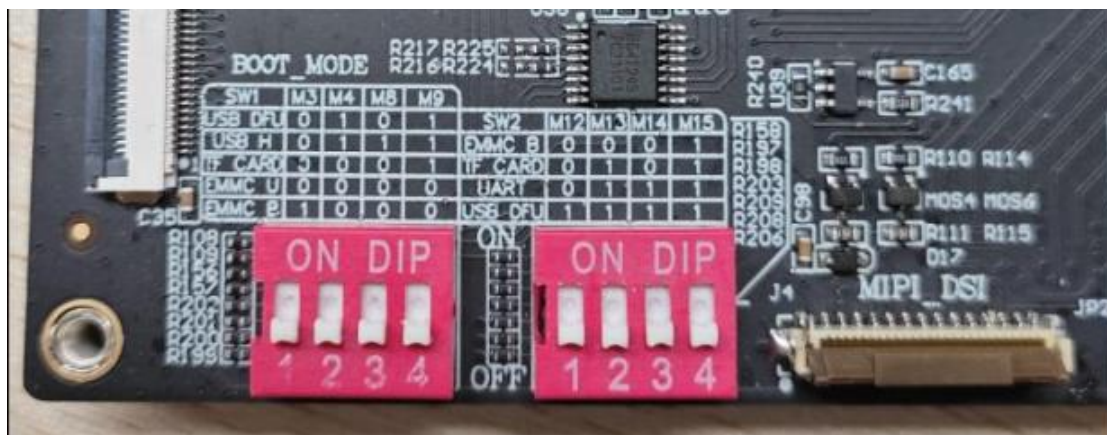


Figure 2.1-3 eMMC Boot (Full Pin Mode)

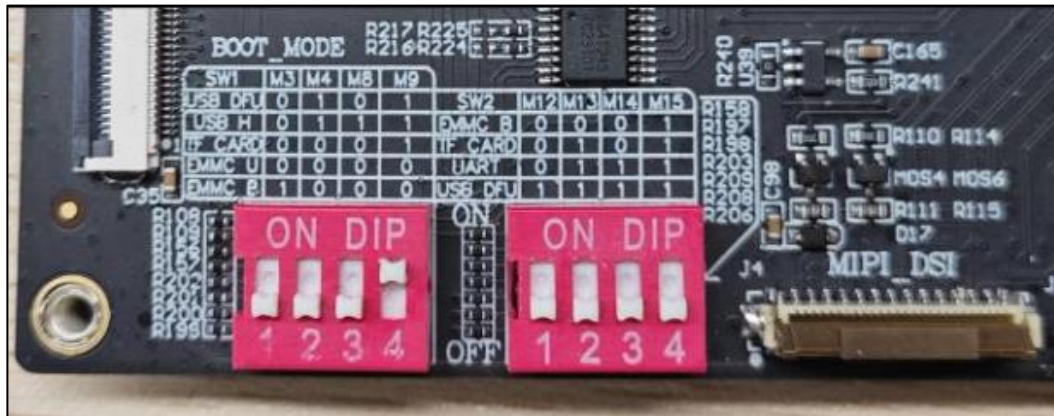


Figure 2.1-4 SD card Boot (Full Pin Mode)

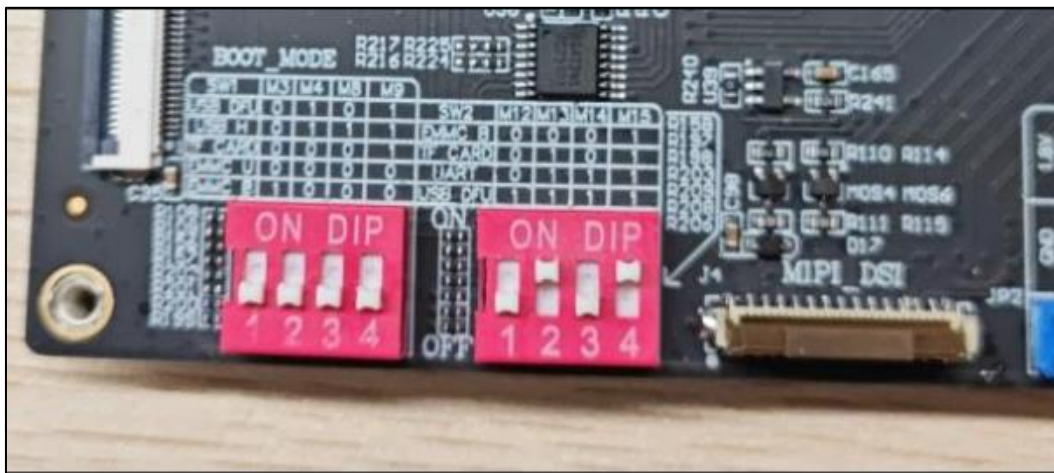


Figure 2.1-5 SD card Boot (Reduce Pin Mode)

2.2 Installing Drivers and Serial Debug Terminal Software Manual

2.2.1 Installing the CH343 Driver

To enable serial communication between the development board and a computer, the CH343 driver must be installed.

On the AM62L Development Board, go to:

Development Board Disc A – Basic Documents → 04_software → CH343SER

Double-click to run the installer. Installation is successful if “Pre-installation succeeded” or “Installation succeeded” appears.

If installation fails on Windows 10, please download Driver Genius.

First, use the supplied USB Type-C cable to connect USB_TTL on the development board baseboard to a USB port on your computer.

Run Driver Genius to scan for hardware drivers, then install the corresponding serial port driver.

2.2.2 How to Use Serial Debug Terminal Software

There are many available serial terminal tools, such as Xshell, SecureCRT, MobaXterm, and even the Ubuntu terminal can connect to the development board’s serial port.

This document uses MobaXterm as an example. It is relatively beginner-friendly among terminal software due to its clean interface, though it has a limitation: it does not support automatic serial reconnection.

On the AM62L Development Board, go to:

Development Board Disc A – Basic Documents → 04_software → MobaXterm_Installer_v24.1.zip

Double-click to extract and install. The installation is straightforward; simply follow the default steps. The installation process is not detailed here.

Once installed, open MobaXterm from the desktop icon. The software interface is shown in the figure below.

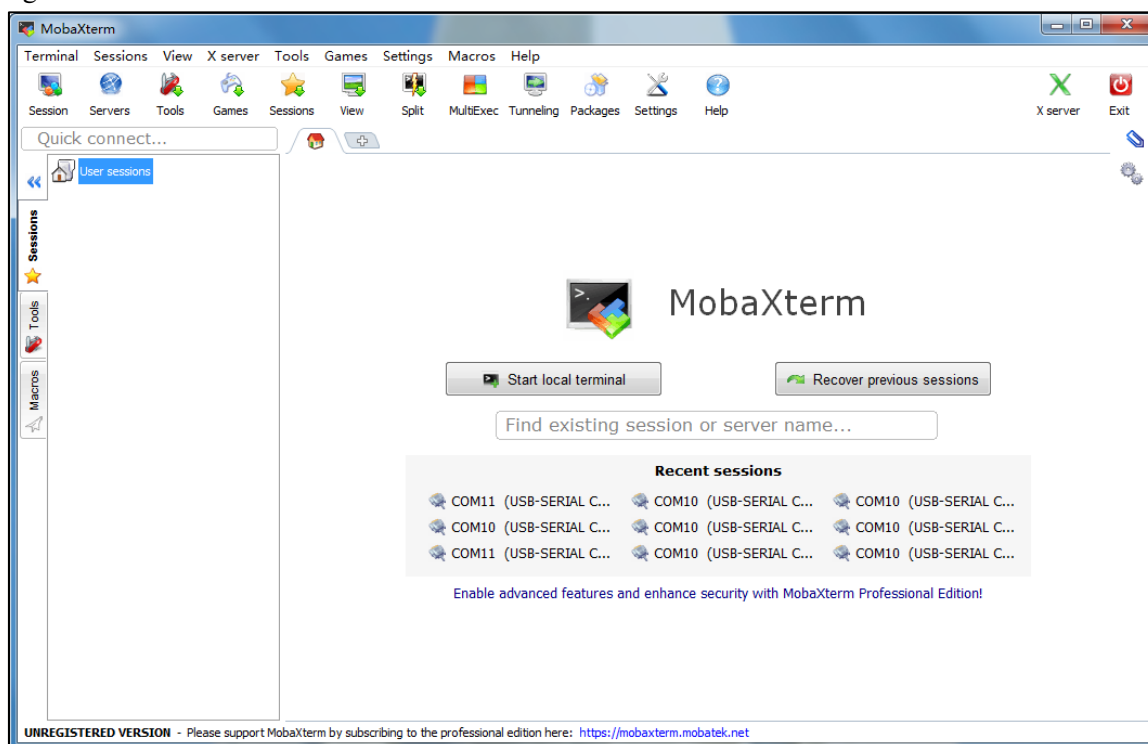


Figure 2.2-1 MobaXterm software interface

Connect the hair board with USB Typec data cable to USB_TTL, connect to the computer, plug in the power supply of the development board, and dial the code switch has been set when shipping. Configure MobaXterm as follows: Click Session→Select Serial type→Select com port corresponding to the development board device→Select Baud rate of 115200 (the default baud rate of the development board is 115200) →Advanced Settings → Set Flow Control as None→Click OK.

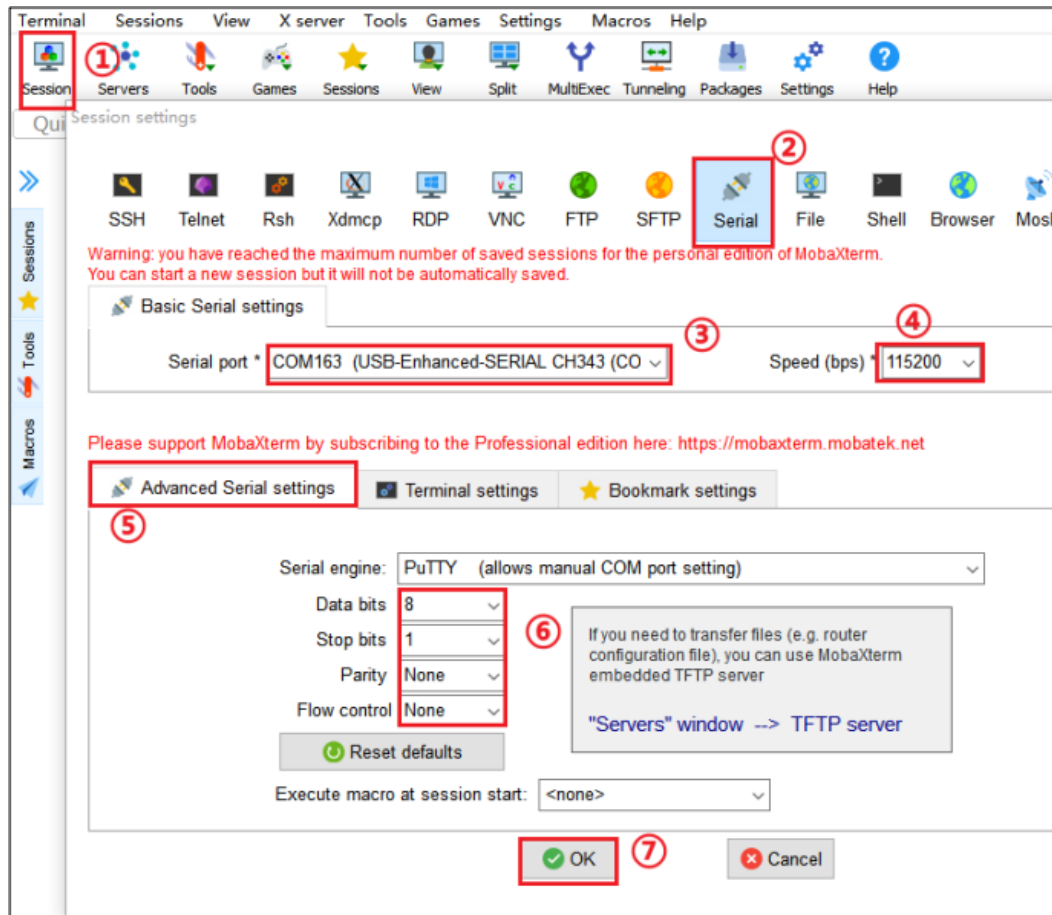


Figure 2.2-2 MobaXterm Operation Steps

After powering on the development board and booting normally, you will see the following print information in the terminal:


```
Starting system message bus: dbus[174]: Unknown username "pulse" in message bus configuration file
done
Starting iptables: OK
Starting bluetoothd: OK
Starting network: [ 5.045736] Bluetooth: MGMT ver 1.23
[ 5.068653] NET: Registered PF_ALG protocol family
[ 5.387606] am65-cpsw-nuss 8000000.ethernet eth0: PHY [8000f00.mdio:01] driver [Generic PHY] (irq=0)
[ 5.397049] am65-cpsw-nuss 8000000.ethernet eth0: configuring for phy/rgmii-rxid link mode
udhcpc: started, v1.36.1
udhcpc: broadcasting discover
udhcpc: no lease, forking to background
[ 8.985792] am65-cpsw-nuss 8000000.ethernet eth1: PHY [8000f00.mdio:02] driver [Generic PHY] (irq=0)
[ 8.995227] am65-cpsw-nuss 8000000.ethernet eth1: configuring for phy/rgmii-rxid link mode
udhcpc: started, v1.36.1
udhcpc: broadcasting discover
udhcpc: no lease, forking to background
OK
Starting connmand: OK
Starting crond: OK
Starting dropbear sshd: [ 12.325682] NET: Registered PF_INET6 protocol family
[ 12.339916] Segment Routing with IPv6
[ 12.343659] In-situ OAM (IOAM) with IPv6
OK
Starting mosquitto: OK
Starting sshd: OK
COLUMNS=99;LINES=52;export COLUMNS LINES;
starting systemui... done.
Starting weston: OK
COLUMNS=99;LINES=52;export COLUMNS LINES;
root@atk-dlam62l:~#
root@atk-dlam62l:~#
```

Figure 2.2-3 Start printing information

Chapter 3. ATK-DLAM62L burning system

This chapter is briefly introduced as follows:

- Using balenaEtcher software
- Burning SD card (the SD card here is burned with a simple file system for eMMC flashing)
- Flashing eMMC

3.1 Factory System Solidification

The official TI AM62L series does not support host computer flashing. Therefore, we use an SD card to boot a small file system, and perform eMMC system solidification under this file system.

AM62L Development Board → Development Board Disc A – Basic Documents → 03. Software → balenaEtcher-Portable-1.18.4.exe

This is a portable software that requires no installation.

(The author has only tested normal use on Windows 10; for other systems, please download from <https://etcher.balena.io/>.)

Double-click to open the file, as shown in the figure below:



Figure 3.1-1 balenaEtcher Interface

The image directory is: AM62L Development Board → Development Board Disc A – Basic Documents → 08. System Images → ATK-DLAM62LB-1g-buildrootV1.0-sd-to-emmc-20260112.img.

Therefore, for users who purchase different versions of the AM62L series, you only need to distinguish between the 512M version, 1G version, and 2G version. Select 512m, 1G, or 2G according to your purchased version. In this example, the author uses the 1G version. Click "Flash from file" to load the corresponding firmware, as shown in the figure below:

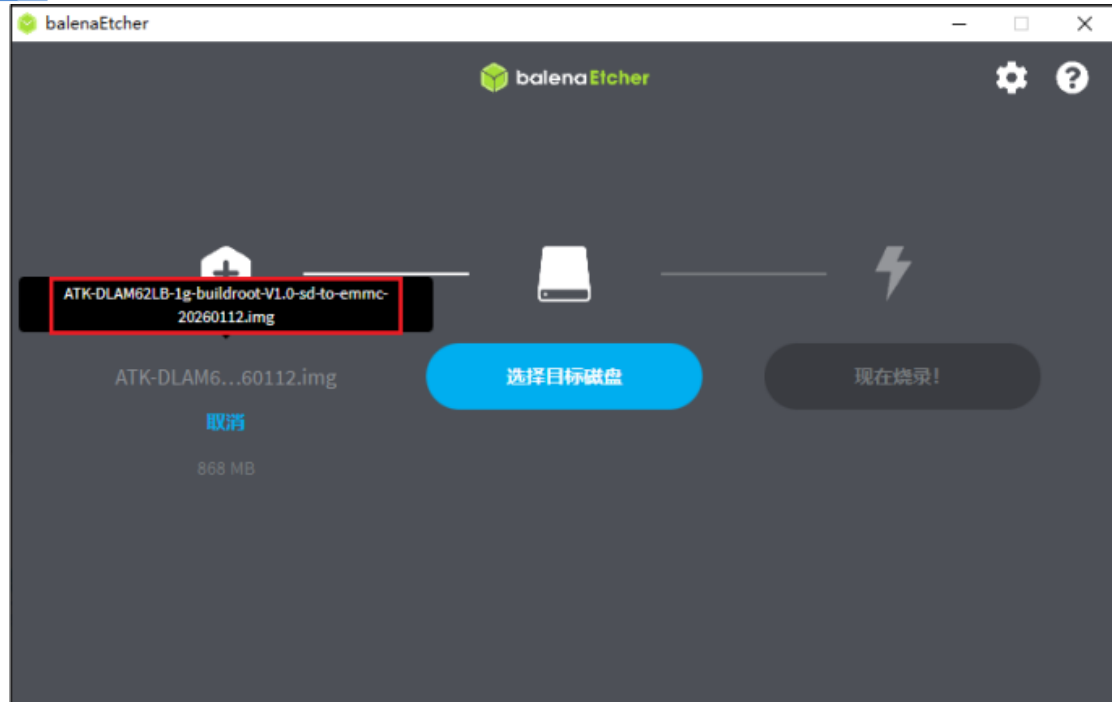


Figure 3.1-2 Select Image



Figure 3.1-3 Card reader

Insert a TF card into the card reader shown above, then connect it to your computer.

Make sure the TF card is recognized by the Windows system.

Click **"Select target disk"**, and your TF card drive will appear, as shown in the figure below:



Figure 3.1-4 Select Disk

You can see the information of our TF card in Step 1. If not, check whether the TF card is mounted under the Ubuntu system or whether the USB port is faulty.

Just follow the steps. Click "**Flash!**" and the following figure will appear:

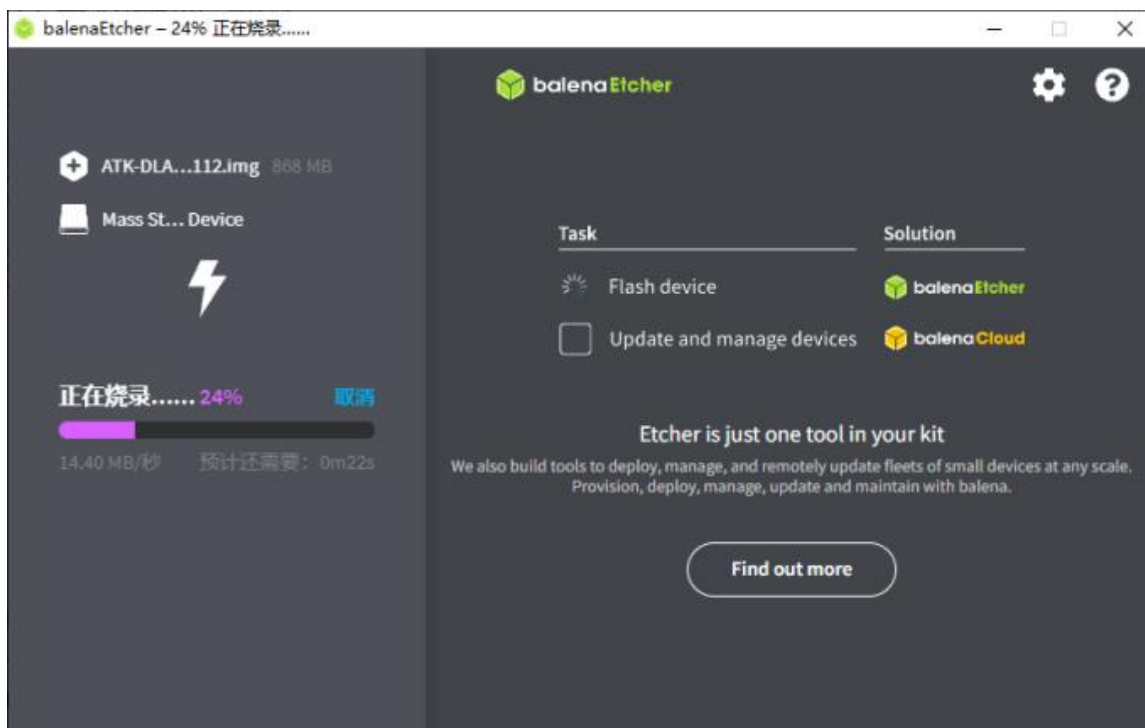


Figure 3.1-5 Flash Image

When the flashing is successful, a Windows formatting window will pop up.

Do NOT click Format. Do NOT click Format. Do NOT click Format.

The successful status is shown in the figure below:

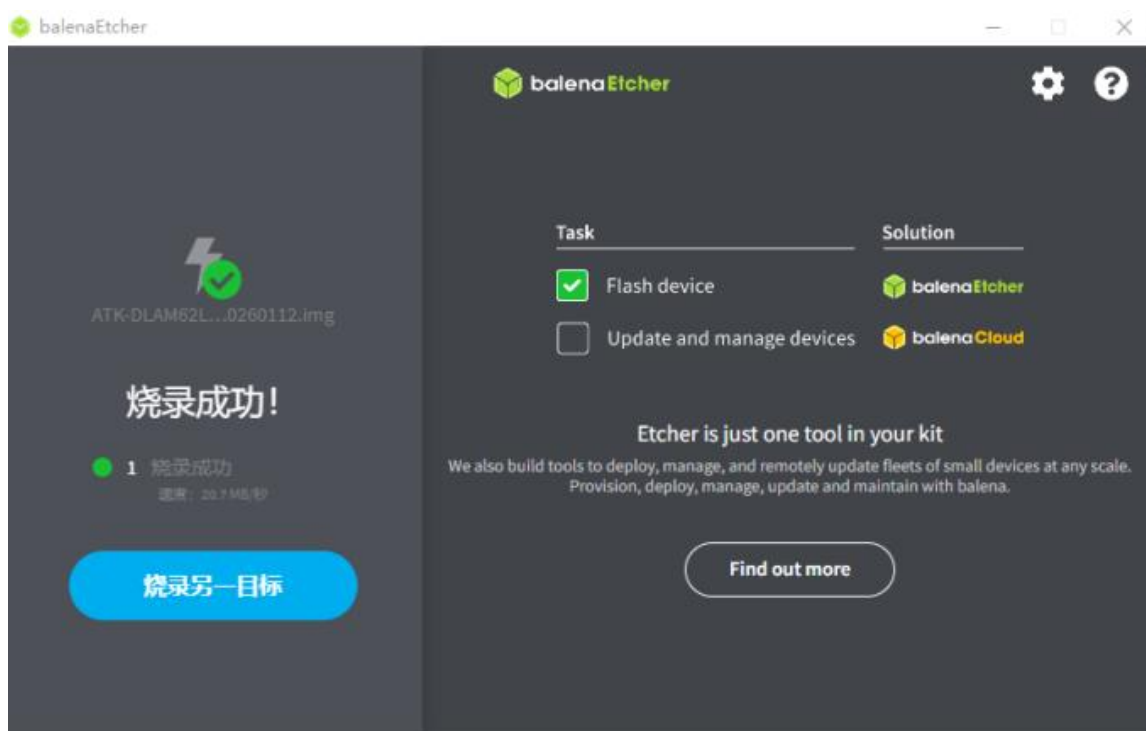


Figure 3.1-6 Flashing completed

Disconnect the card reader from Windows, then insert the TF card into the ATK-DLAM62L development board, as shown in the figure below:

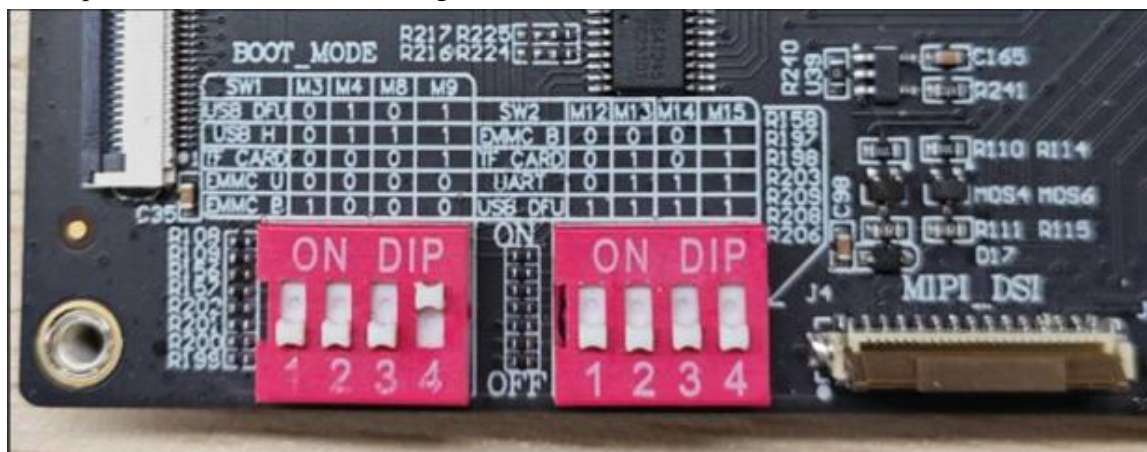


Figure 3.1-7 Boot Mode Selection

The DIP switches in the figure above are set for SD card boot. For detailed boot methods, refer to Section 2.1.

Connect the USB TTL serial port, and the serial terminal will display the following print information:

```

Hardware is initialized using a generic method
No state is present for card ES8390
FAIL
Starting network: OK
# [ 2.961325] usb 1-1.3: new high-speed USB device number 3 using xhci-hcd
[ 3.772652] mmcblk0: p1 p2
格式化分区...
[ 8.177896] EXT4-fs (mmcblk0p2): mounted filesystem 17a78c98-1dd2-11b2-8792-f70ea860114e r/w with ordered data mode. Quota mode: none.
解压根文件系统...
拷贝文件系统...
同步数据...
卸载分区...
[ 77.730884] EXT4-fs (mmcblk0p2): unmounting filesystem 17a78c98-1dd2-11b2-8792-f70ea860114e.
Simple mixer control 'DACL',0
Capabilities: pvolume pvolume-joined
Playback channels: Mono
Limits: Playback 0 - 255
Mono: Playback 155 [61%] [-10.00dB]
Simple mixer control 'DACR',0
Capabilities: pvolume pvolume-joined
Playback channels: Mono
Limits: Playback 0 - 255
Mono: Playback 155 [61%] [-10.00dB]

```

Figure 3.1-8 Flashing Completed

You can see the red LED on the development board is flashing. It will stop flashing and stay solid on once the flashing is finished. At the same time, the speaker will play a tone.

You can then power off the board and set the DIP switches to "0000 0000" for eMMC boot.

```

U-Boot SPL 2025.01-g606317532dc7 (Jan 12 2026 - 11:13:14 +0800)
CPL initial stack usage: 1872 bytes
Trying to boot from MMC1
ERROR: Agent 0 Protocol 0x10 Message 0x7: not supported

U-Boot 2025.01-g606317532dc7 (Jan 12 2026 - 11:13:14 +0800)

SoC: AM62LX SR2.0 HS-FS
Model: Texas Instruments AM62L3 Evaluation Module
DRAM: 1 GiB
ERROR: Agent 0 Protocol 0x10 Message 0x7: not supported
Core: 80 devices, 31 uclasses, devicetree: separate
MMC: mmc@fa10000: 0, mmc@fa00000: 1
Loading Environment from nowhere... OK
In: serial@2800000

```

Figure 3.1-9 Boot Media

The initial printout indicates that the boot device is MMC1, which refers to eMMC boot.

Chapter 4. Functional test

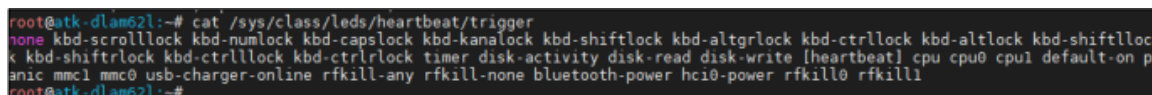
4.1 LED Testing

The ATK-DLAM62L development board is equipped with three LEDs: one blue system power indicator, one red user LED, and one yellow-green 4G status indicator.

In the factory system, enter the following commands to control the user LED (DS1) and view the current trigger mode of the DS1 LED on the system.

```
cat /sys/class/leds/heartbeat/trigger // See the current and supported triggers for DS1
```

The result is shown in the following screenshot:



```
root@atk-dlam62l:~# cat /sys/class/leds/heartbeat/trigger
none kbd-scrolllock kbd-numlock kbd-capslock kbd-kana-lock kbd-shiftlock kbd-altgrlock kbd-ctrllock kbd-altlock kbd-shiftilloc
k kbd-shiftlock kbd-ctrllock kbd-ctrllock timer disk-activity disk-read disk-write [heartbeat] cpu cpu0 cpu1 default-on p
anic mmc1 mmc0 usb-charger-online rkill-any rkill-none bluetooth-power hci0-power rkill0 rkill1
root@atk-dlam62l:~#
```

Figure 4.1-1 View trigger mode

It can be seen that the trigger mode of DS1 at this time is [heartbeat], which means that the current trigger condition is the system heartbeat light. The trigger mode can be modified by modifying the string of trigger. The command will look as follows:

```
echo none > /sys/class/leds/heartbeat/trigger // Change how DS1 is triggered, setting it to none
```

The following command is used to control the DS1 lamp.

```
echo 1 > /sys/class/leds/heartbeat/brightness //Light up DS1
```

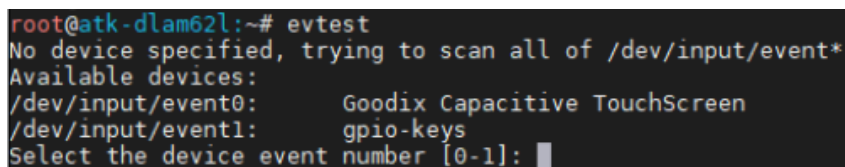
```
echo 0 > /sys/class/leds/heartbeat/brightness //Put out DS1
```

4.2 KEY test

The ATK-DLAM62L development board is equipped with four keys, including two reset keys, one WKUP key, and one user key.

In the factory system, enter the following commands to control the user key. The commands are as follows:

```
evtest
```



```
root@atk-dlam62l:~# evtest
No device specified, trying to scan all of /dev/input/event*
Available devices:
/dev/input/event0:      Goodix Capacitive TouchScreen
/dev/input/event1:      gpio-keys
Select the device event number [0-1]: █
```

Figure 4.2-1 evtest interface

Figure 4.2.1 tells us that "/dev/input/event1" is the interrupt event of gpio-keyboard, just enter 1 (if there are multiple input devices, the number may change, you need to enter the number according to the actual situation)

```

Select the device event number [0-1]: 1
Input driver version is 1.0.1
Input device ID: bus 0x19 vendor 0x1 product 0x1 version 0x100
Input device name: "gpio-keys"
Supported events:
  Event type 0 (EV_SYN)
  Event type 1 (EV_KEY)
    Event code 256 (BTN_0)
Key repeat handling:
  Repeat type 20 (EV_REP)
  Repeat code 0 (REP_DELAY)
    Value 250
  Repeat code 1 (REP_PERIOD)
    Value 33
Properties:
Testing ... (interrupt to exit)

```

Figure 4.2-2 Keypress value

In Figure 4.2.2, the red box represents the key value reported by the key. Printing information from the kernel appears at the press of a key. As shown below:

```

Properties:
Testing ... (interrupt to exit)
Event: time 140.627057, type 1 (EV_KEY), code 256 (BTN_0), value 1
Event: time 140.627057, ----- SYN_REPORT -----
Event: time 140.842627, type 1 (EV_KEY), code 256 (BTN_0), value 0
Event: time 140.842627, ----- SYN_REPORT -----
Event: time 141.149087, type 1 (EV_KEY), code 256 (BTN_0), value 1
Event: time 141.149087, ----- SYN_REPORT -----
Event: time 141.329290, type 1 (EV_KEY), code 256 (BTN_0), value 0
Event: time 141.329290, ----- SYN_REPORT -----
Event: time 141.949111, type 1 (EV_KEY), code 256 (BTN_0), value 1
Event: time 141.949111, ----- SYN_REPORT -----
Event: time 142.122780, type 1 (EV_KEY), code 256 (BTN_0), value 0
Event: time 142.122780, ----- SYN_REPORT -----

```

Figure 4.2-3 Key press test

4.3 LCD Testing

Only kernel display is shown here; there are no relevant configurations for U-Boot.

The ATK-DLAM62L development board supports display screens with RGB and MIPI interfaces (MIPI and RGB share the same group of IOs and can only be multiplexed as one type of display interface).

The factory system defaults to the 720P screen with MIPI interface, using the device tree `alientek-am62l-mipi-5p5-720p.dtbo` (dtbo can be understood as an extended device tree, loaded via the `uEnv.txt` file).

RGB screen device tree	MIPI screen device tree
alientek-am62l-rgb-4p3-272p.dtbo	alientek-am62l-mipi-5p5-720p.dtbo
alientek-am62l-rgb-4p3-480p.dtbo	alientek-am62l-mipi-5p5-1080p.dtbo
alientek-am62l-rgb-7-480p.dtbo	alientek-am62l-mipi-10p1-800p.dtbo
alientek-am62l-rgb-7-600p.dtbo	
alientek-am62l-rgb-10p1-800p.dtbo	

Judging from their suffixes, they are all dtbo files, and these screen-related device trees are all overlay device trees.

4.3.1 RGB Interface Test Manual

Note: Do not connect a MIPI screen. You can only choose either RGB or MIPI screen.

This section requires basic Linux knowledge, and proficiency with the vim or vi editor is necessary. If you are not familiar with them, please learn by yourself via Baidu.

The factory system defaults to a 720P screen using the MIPI interface. If you need to switch to an RGB screen display, you can modify the uEnv.txt file.

This test uses a 7-inch RGB screen (1024x600). The hardware connection method is shown in the figure below:

(You can flip the black buckle upward, insert the screen ribbon cable with the blue side facing up, then press the buckle down.)

As shown in the figure:

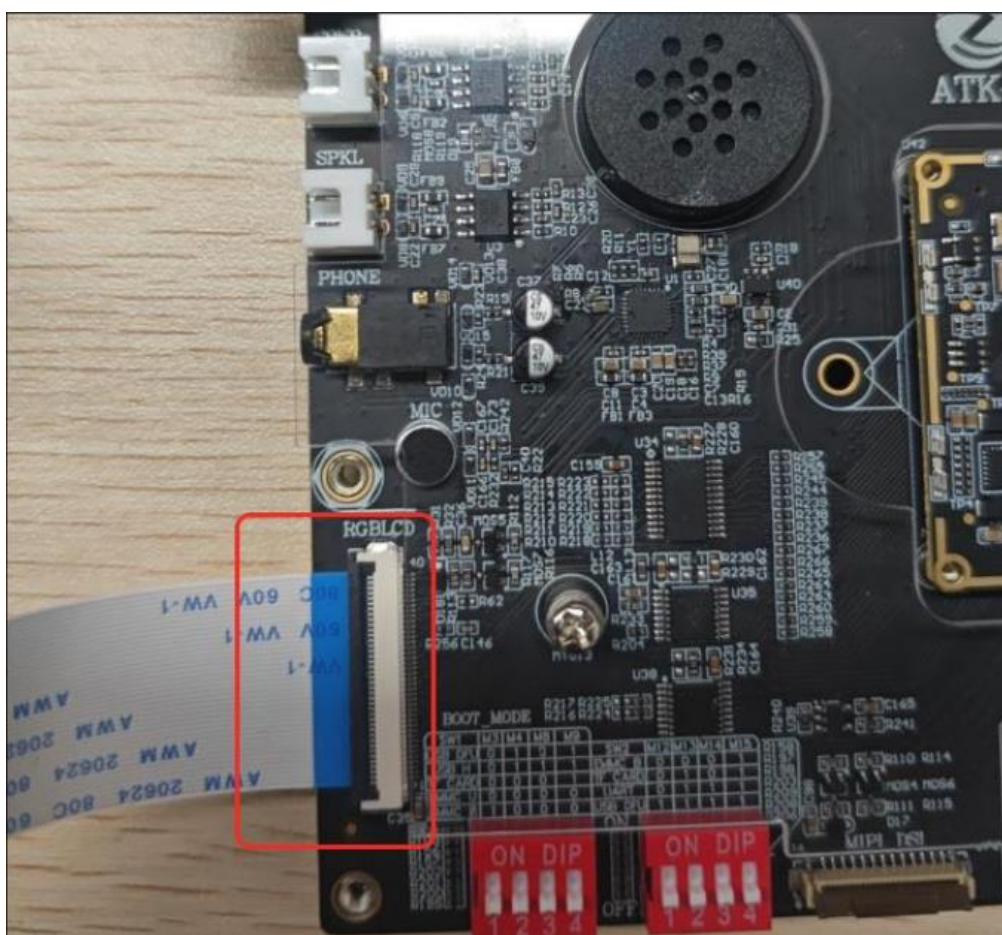


Figure 4.3-1 RGB interface

After connecting the screen, plug in the debug serial port (USB TTL). We will power on the development board first; the author uses eMMC boot here. Once the boot is complete, run the following code to modify the screen configuration:

```
vi /boot/uEnv.txt
```

You can see the following code segment:

```
1 # This uEnv.txt file can contain additional environment settings that you
2 # want to set in U-Boot at boot time. This can be simple variables such
3 # as the serverip or custom variables. The format of this file is:
```

```

4 # variable=value 手册
5 # NOTE: This file will be evaluated after the bootcmd is run and the
6 # bootcmd must be set to load this file if it exists (this is the
7 # default on all newer U-Boot images. This also means that some
8 # variables such as bootdelay cannot be changed by this file since
9 # it is not evaluated until the bootcmd is run.
10
11 #name_overlays=ti/alientek-am62l-mipi-10p1-800p.dtbo
12 #name_overlays=ti/alientek-am62l-mipi-5p5-1080p.dtbo
13 name_overlays=ti/alientek-am62l-mipi-5p5-720p.dtbo
14
15 #name_overlays=ti/alientek-am62l-rgb-10p1-800p.dtbo
16 #name_overlays=ti/alientek-am62l-rgb-4p3-272p.dtbo
17 #name_overlays=ti/alientek-am62l-rgb-4p3-480p.dtbo
18 #name_overlays=ti/alientek-am62l-rgb-7-480p.dtbo
19 #name_overlays=ti/alientek-am62l-rgb-7-600p.dtbo

```

uEnv.txt is the environment variable configuration file for the U-Boot bootloader. After adding or modifying environment variable settings in this file, U-Boot will read the uEnv.txt file during the boot process and update the corresponding environment variables according to its contents, thereby overriding the original default or configured parameters.

The # character is used for comments.

Line 13 is uncommented, indicating that we define the name_overlays environment variable and set it to ti/alientek-am62l-mipi-5p5-720p.dtbo, which the system loads as the device tree.

To use the 7-inch 1024x600 RGB screen, comment out Line 13 and enable Line 19.

After modification, it is shown in the figure below:



Figure 4.3-2 uEnv.txt

After completing the modifications, simply reboot the development board.

You can verify the configuration by checking whether the screen lights up, or by running the following command via the board's serial debug console:

```
fbset
```



```

root@atk-dlam62l:~# fbset

mode "1024x600-0"
# D: 0.000 MHz, H: 0.000 kHz, V: 0.000 Hz
geometry 1024 600 1024 600 32
timings 0 0 0 0 0 0
accel false
rgba 8/16,8/8,8/0,0/0
endmode

root@atk-dlam62l:~#

```

Figure 4.3-3 fbset

From the information in the figure above, we can confirm that our screen has a resolution of 1024x600.

4.3.2 MIPI DSI Interface Test

The factory system defaults to a 720P screen using the MIPI interface. If you need to switch to a MIPI screen with a different resolution, please replace it according to the connection method shown in the figure below:

(Flick the brown buckle upward to release it and remove the ribbon cable; press it downward to lock the buckle and secure the ribbon cable.)



Figure 4.3-4 MIPI DSI interface

The modification method is the same as for RGB, you just load a different screen device tree via uEnv.txt.

4.3.3 Touch Test

The touch test is simple and uses the evtest command just like the key test. Enter the following command in the debug terminal of the development board:

```
evtest
```

```

root@atk-dlam62l:~# evtest
No device specified, trying to scan all of /dev/input/event*
Available devices:
/dev/input/event0:    Goodix Capacitive TouchScreen
/dev/input/event1:    gpio-keys
Select the device event number [0-1]: 

```

Figure 4.3-5 evtest tool

/dev/input/event0 is the screen touch test device. Enter 0 directly to test the screen touch.

4.3.4 Backlight Test

In the default factory system, RGB, LVDS and MIPI screen backlight support 256 levels of PWM adjustment, brightness level 0 to 255, the default 255. When set to 0, the backlight is turned off. The higher the number, the brighter the screen. The RGB screen test code is as follows:

```
cat /sys/class/backlight/backlight-dsi/brightness // Check the LCD maximum brightness level
echo 100 > /sys/class/backlight/backlight-dsi/brightness // Modify the current brightness level and
observe the brightness changes on the screen.
```

The code below is for the RGB screen backlight adjustment test:

```
cat /sys/class/backlight/backlight-rgb/brightness // Check the current brightness level
echo 100 > /sys/class/backlight/backlight-rgb/brightness // Modify the current brightness level and
observe the changes in screen brightness.
```

4.4 Serial port test

ATK-DLAM62L development board has two RS485 (uart1 and uart4) on board. Here, we need to use the ALIENTEK USB converter module to test. If the user has 485 related test tools to test by themselves, it is not necessary to use this module. Figure below.



Figure 4.4-1 ALIENTEK multi-in-one USB converter module

Connect the A of RS485 of the base board to the A of the ALIENTEK USB converter module with dupont wire, and connect B to B. Using the ALIENTEK USB converter module, you will see a serial port in the PC(computer side). The hardware connection is shown in the following figure:



Figure 4.4-2 Connection mode between development board and serial port tool

uart1 corresponds to ttyS3 and uart4 corresponds to ttyS6.

Configure the serial port using the following commands.

```
stty -F /dev/ttyS3 3000000 icanon iexten -ixon -ixoff -opost -echo min 1 time 0
```

The stty instruction is explained:

(1) -F (--file): Open the specified device and use this device as the input instead of the standard input.

(2) 3000000: Set the serial port baud rate to 3000000bps (3Mbps);

(3) icanon: Enable canonical mode, also known as line-buffering mode;

(4) iexten: Enable extended input processing.

For more usage instructions of this command, enter stty -help.

In this test, **XCOM V2.0** is used as the host computer to communicate with the development board.

The software path is: AM62L Development Board → Development Board Disc A - Basic Resources → 04. Software → XCOM V2.0.

Enter the following commands to wait for data sent from the host computer.

Note: Do not send Chinese characters, otherwise garbled characters may occur.

Press Ctrl + C to stop receiving.

```
cat /dev/ttyS3
```

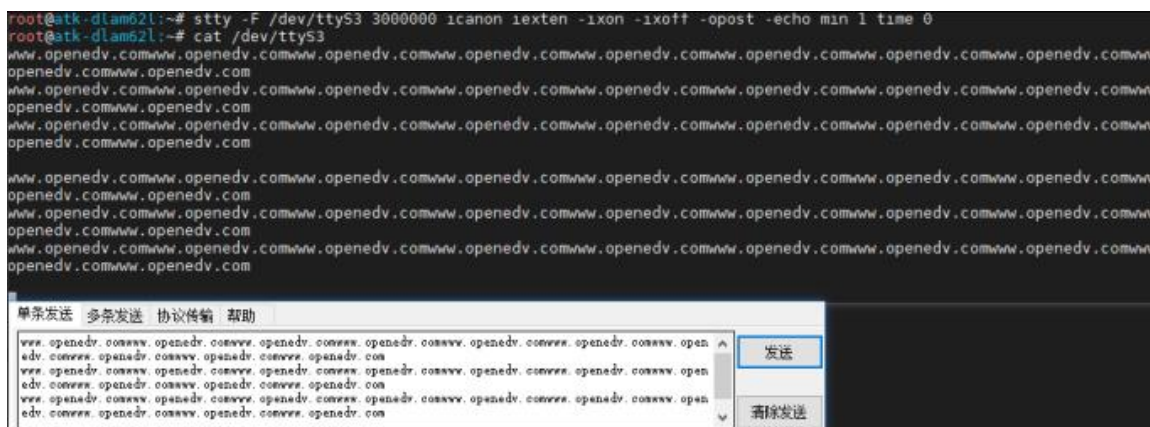


Figure 4.4-3 The serial port receives data

Receive the 'cat' program, and use the following command to send data to the host computer:

echo

www.openedv.comwww.openedv.comwww.openedv.comwww.openedv.comwww.openedv.comw

ww.openedv.com www.openedv.comwww.openedv.comwww.openedv.comwww.openedv.com >
/dev/ttyS3

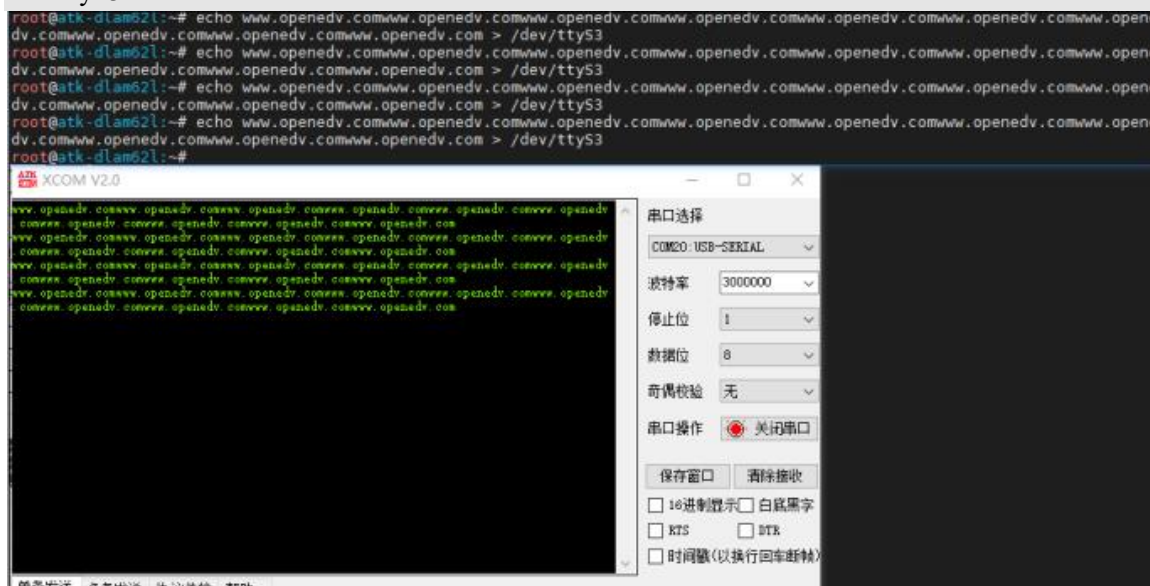


Figure 4.4-4 Serial port to send data

4.5 USB Testing

The AM62L chip has two usb ports: USB HOST and USB OTG. ATK-DLAM62L development board uses CH334R to transfer USB HOST out of 4 USB HOST interfaces (two as ordinary USB HOST, one as 4G module, one as usb WiFi). USB OTG is directly exported as the OTG interface of the development board.

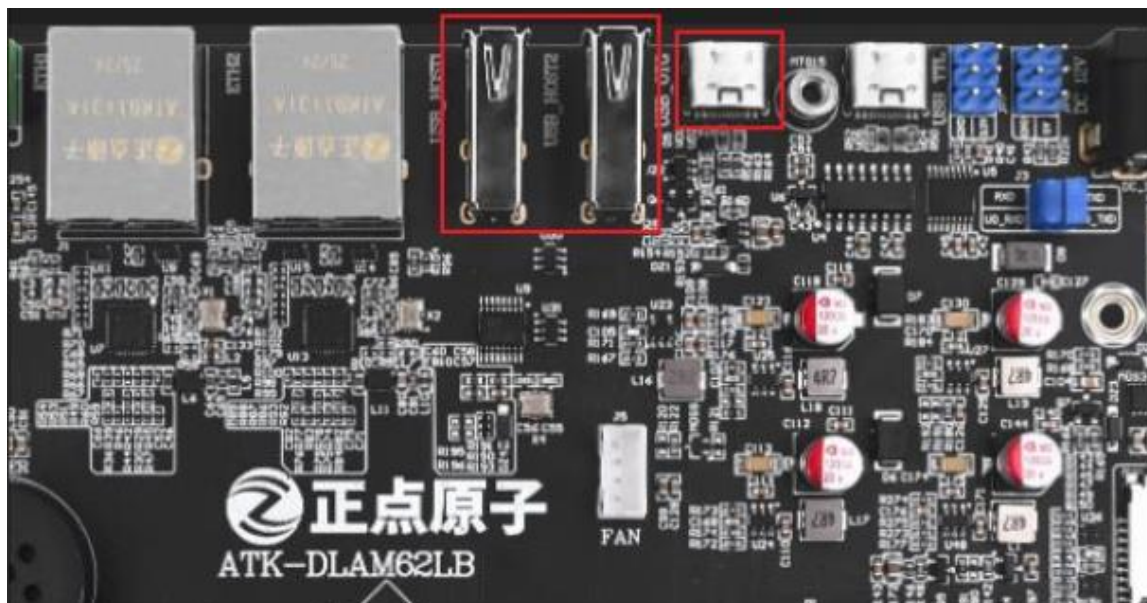


Figure 4.5-1 USB interface

4.5.1 USB HOST Test

Test requirements: Get a USB disk (FAT32 format). The development board is normally powered on and connected to the debugging serial port. When we connect the USB disk to the USB HOST interface, the following message is printed:

```

root@ATK-DLT113IS:/# [ 132.895406] usb 1-1.2: new high-speed USB device number 5 using sunxi-ehci
[ 133.047986] usb-storage 1-1.2:1.0: USB Mass Storage device detected
[ 133.055925] scsi host0: usb-storage 1-1.2:1.0
[ 134.126263] scsi 0:0:0:0: Direct-Access      Mass Storage Device 1.00 PQ: 0 ANST: 0 CCS
[ 134.267371] sd 0:0:0:0: [sda] 60579840 512-byte logical blocks: (31.0 GB/28.9 GiB)
[ 134.276601] sd 0:0:0:0: [sda] Write Protect is off
[ 134.282065] sd 0:0:0:0: [sda] Mode Sense: 03 00 00 00
[ 134.288607] sd 0:0:0:0: [sda] No Caching mode page found
[ 134.294598] sd 0:0:0:0: [sda] Assuming drive cache: write through
[ 134.344614] sda: sda1
[ 134.357825] sd 0:0:0:0: [sda] Attached SCSI removable disk
create /dev/sda
create /dev/sda1
[ 134.444134] FAT-fs (sda1): Volume was not properly unmounted. Some data may be corrupt. Please run fsck.
  
```

The capacity of the U disk
sda denotes the node of the USB drive,
and sda1 denotes that the USB drive
has only one partition, which is sda1

Figure 4.5-2 USB initialization information

See the above printed information, indicating that the U disk identification is successful. Here the author's U disk format for FAT32 so, can be automatically mounted. To see if the mount was successful, run the following command:

```
df -h
```

```

root@atk-dlam62l:~#
root@atk-dlam62l:~# df -h
Filesystem      Size      Used Available Use% Mounted on
/dev/root        28.1G    558.6M    26.2G    2% /
devtmpfs         469.1M         0    469.1M    0% /dev
tmpfs            486.5M         0    486.5M    0% /dev/shm
tmpfs            486.5M    68.0K    486.4M    0% /tmp
tmpfs            486.5M    208.0K    486.3M    0% /run
/dev/mmcblk1p1   126.0M    58.3M    67.7M   46% /boot
root@atk-dlam62l:~# mount /dev/sd1 /mnt/
root@atk-dlam62l:~# df -h
Filesystem      Size      Used Available Use% Mounted on
/dev/root        28.1G    558.6M    26.2G    2% /
devtmpfs         469.1M         0    469.1M    0% /dev
tmpfs            486.5M         0    486.5M    0% /dev/shm
tmpfs            486.5M    68.0K    486.4M    0% /tmp
tmpfs            486.5M    208.0K    486.3M    0% /run
/dev/mmcblk1p1   126.0M    58.3M    67.7M   46% /boot
/dev/sd1         29.5G   128.0K    29.5G    0% /mnt
root@atk-dlam62l:~#

```

Figure 4.5-3 USB device mount information

4.5.2 USB OTG testing

- OTG HOST test

See Section 4.5.1 for the test method. A typeC to typeA master line is required. The following image:



Figure 4.5-4 typeC to typeA master line

- OTG slave test

The OTG slave device is configured as an ADB function in the default factory system.

Testing is performed under the Ubuntu system. Run the following commands in Ubuntu to install ADB.

```
sudo apt install adb
```

Execute the ADB initialization script on the development board. An ADB device will be generated after initialization. This function does not support hot-plugging.

If the OTG cable is unplugged or the board is rebooted, the script must be run again. The command is as follows:

```
/opt/shell/usb_adb
```



```

root@atk-dlam62l:~# ./opt/shell/usb_adb
Creating gadget...
[ 48.250774] file system registered
Starting adbd...
Waiting for adbd to be ready...
install_listener('tcp:5037','*smartsocket*')[ 48.784218] read descriptors

[ 48.790323] read strings
adbd ready (ep1 found)
USB ADB gadget started on 31000000.usb
Done. Now run 'adb devices' on PC.

```

Figure 4.5-5 Start the ADB service.

Connect the development board to the computer using the Type-C interface. A device named "Google Configsffs gadget" will appear under the mobile device list in VMware. Connect this device to the virtual machine. A successful connection is shown in the figure below:

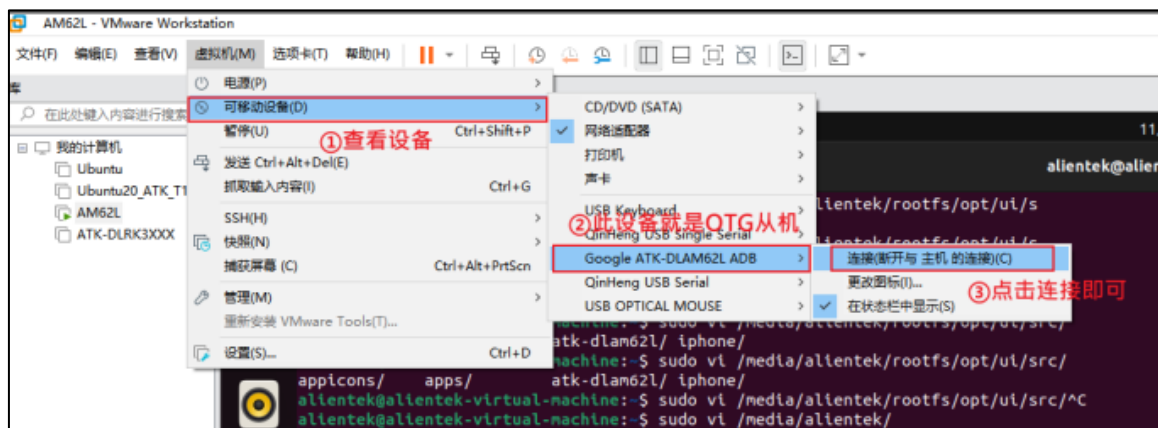


Figure 4.5-6 Connecting to a virtual machine

ADB commands are commonly used in three: shell, push and pull, other advanced commands please do your own research.

1. shell

```

adb shell
exit // Exit terminal

```

Enter the terminal of the development board, and the terminal at this time is relative to the debugging serial port. As shown below:

```

alientek@alientek-virtual-machine:~$ adb devices
* daemon not running; starting now at tcp:5037
* daemon started successfully
List of devices attached
123456 device

alientek@alientek-virtual-machine:~$ adb shell
root@atk-dlam62l:~#
root@atk-dlam62l:~#
root@atk-dlam62l:~#

```

Figure 4.5-7 adb shell

1. push

```
adb push xxxx /
```

push means to copy files to the board. Here "xxxx" means the file to copy, and "/" means copy to the root directory.

2. pull

```
adb pull xxxx ./
```

pull means to copy the file of the development board to the Ubuntu system, "xxxx" means to copy the file, "./" means to copy to the current directory of the Ubuntu system.

4.6 Network testing

ATK-DLT113IS development board equipped with a gigabit network chip, adaptive 10/100/1000M. Use the following command to test the network in the filesystem, and plug the network cable into the network port to ensure that the network cable can access the Internet. Check the IP obtained by the network.

```
ifconfig
```

```
root@atk-dlam62l:~# ifconfig
eth0      Link encap:Ethernet  HWaddr 30:30:D0:97:26:CE
          inet addr:192.168.6.224  Bcast:192.168.6.255  Mask:255.255.255.0
          inet6 addr: fe80::3230:d0ff:fe97:26ce/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:153  errors:0  dropped:33  overruns:0  frame:0
          TX packets:31  errors:0  dropped:95  overruns:0  carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:12471 (12.1 KiB)  TX bytes:4721 (4.6 KiB)

eth1      Link encap:Ethernet  HWaddr A6:E9:63:D7:CD:70
          UP BROADCAST MULTICAST  MTU:1500  Metric:1
          RX packets:0  errors:0  dropped:0  overruns:0  frame:0
          TX packets:0  errors:0  dropped:98  overruns:0  carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)
```

Figure 4.6-1 Look at the enabled network node eth0

Check whether the network can access the Internet, if there is a data reply, it means that the network function is normal. Press Ctrl+C to terminate the command.

```
ping -I eth0 www.baidu.com // Use eth0 network port ping Baidu
```

```
root@atk-dlam62l:~# ping -I eth0 www.baidu.com
PING www.baidu.com (183.2.172.177): 56 data bytes
64 bytes from 183.2.172.177: seq=0 ttl=52 time=3.784 ms
64 bytes from 183.2.172.177: seq=1 ttl=52 time=3.800 ms
64 bytes from 183.2.172.177: seq=2 ttl=52 time=2.849 ms
64 bytes from 183.2.172.177: seq=3 ttl=52 time=2.993 ms
64 bytes from 183.2.172.177: seq=4 ttl=52 time=3.743 ms
^C
--- www.baidu.com ping statistics ---
5 packets transmitted, 5 packets received, 0% packet loss
round-trip min/avg/max = 2.849/3.433/3.800 ms
root@atk-dlam62l:~#
```

Figure 4.6-2 ping Baidu

Test whether it is Gigabit network, pay attention to the use of gigabit network cable, gigabit router or gigabit switch, PC (computer) network card must be gigabit network card, or the speed of the test or recognition may be 100 megabit! In the figure below, you can see that the software identifies 1Gbps/Full as a gigabit network.

```
root@atk-dlam62l:~# [ 539.790262] adbd (504): /proc/504/oom_adj is deprecated, please use /proc/504/oom_score_adj instead.
[ 582.200658] am65-cpsw-nuss 8000000.ethernet eth0: Link is Up 1Gbps/Full flow control rx/tx
```

Figure 4.6-3 Identified as a gigabit network

Gigabit network connection speed can be tested using the iperf3 command.

Set up Ubuntu as the server.

iperf3 -s

ATK-DLAM62L the client, through the iperf3 command, enter the ip of the server.

iperf3 -B 192.168.6.224 -c 192.168.6.205 -i 1

Explanation:

- -B 192.168.6.224: IP address of the client, the development board has two IP addresses, one address needs to be specified. Or default to the eth0 network port.
- -c 192.168.6.205: The IP address of the Ubuntu server.
- -i 1: Trigger period in seconds.

```
root@atk-dlam62l:~# iperf3 -B 192.168.6.224 -c 192.168.6.205 -i 1
Connecting to host 192.168.6.205, port 5201
[ 5] local 192.168.6.224 port 34061 connected to 192.168.6.205 port 5201
[ ID] Interval          Transfer      Bitrate      Retr  Cwnd
[ 5] 0.00-1.00 sec      111 MBytes   932 Mbits/sec  0     553 KBytes
[ 5] 1.00-2.00 sec      108 MBytes   907 Mbits/sec  0     581 KBytes
[ 5] 2.00-3.00 sec      108 MBytes   904 Mbits/sec  0     581 KBytes
[ 5] 3.00-4.00 sec      108 MBytes   905 Mbits/sec  0     581 KBytes
[ 5] 4.00-5.00 sec      108 MBytes   907 Mbits/sec  0     581 KBytes
[ 5] 5.00-6.00 sec      108 MBytes   905 Mbits/sec  0     581 KBytes
[ 5] 6.00-7.00 sec      109 MBytes   918 Mbits/sec  0     611 KBytes
[ 5] 7.00-8.00 sec      109 MBytes   911 Mbits/sec  0     643 KBytes
[ 5] 8.00-9.00 sec      108 MBytes   905 Mbits/sec  0     643 KBytes
[ 5] 9.00-10.00 sec     108 MBytes   905 Mbits/sec  0     643 KBytes
- - - - -
[ ID] Interval          Transfer      Bitrate      Retr
[ 5] 0.00-10.00 sec     1.06 GBytes   910 Mbits/sec  0
[ 5] 0.00-10.05 sec     1.06 GBytes   904 Mbits/sec
iperf Done.
root@atk-dlam62l:~#
```

Figure 4.6-4 Gigabit network testing

If the gigabit network speed is not measured, please check whether the gigabit network cable is used, whether the network card of the PC is gigabit network card, whether the router is gigabit, etc., can not have a 100 megabit device, otherwise the test is 100 megabit network.

Note: The actual test can not run the theoretical value, please refer to the actual test speed.

4.7 CAN Test

There are two channels of CAN on the ATK-DLAM62L development board. To test the CAN communication function, users need to have an instrument to test CAN (can also use two channels of CAN docking communication), such as Zhou Ligong's CAN analyzer, Chuangxin Technology's CAN analyzer and Guangcheng Technology's CAN analyzer. For the use of CAN instrument and CAN host computer, please refer to the manufacturer's manual. If you CAN't use it, please consult the CAN manufacturer's technical support.

The main CAN features of ATK-DLAM62L development board are as follows:

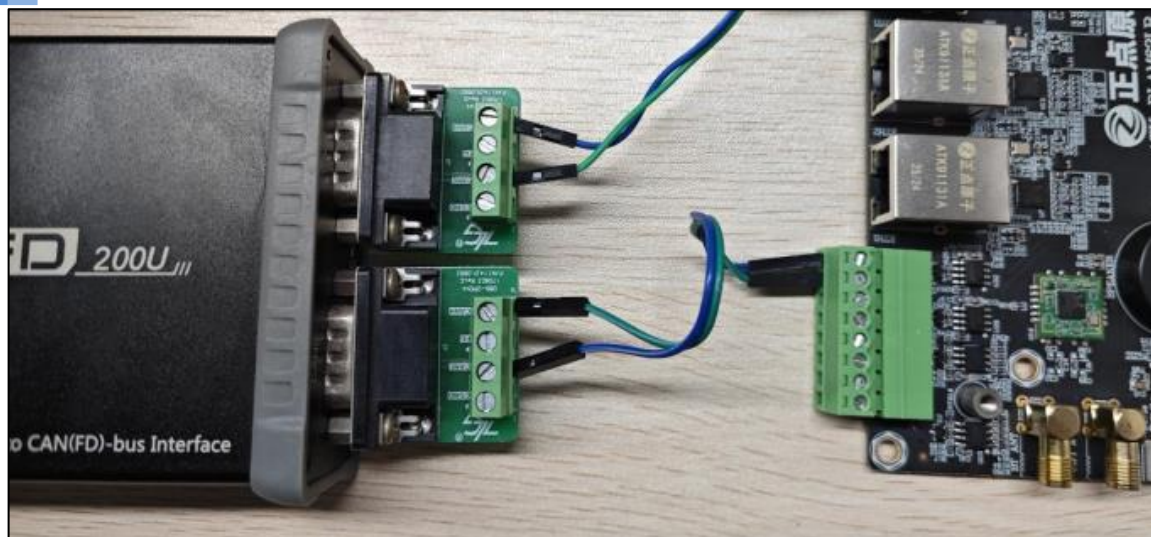


Figure 4.7-1 USBCANFD-200U

4.7.1 CAN test

Use the following commands for testing. First, enable the CAN device and configure the baud rate:

```
ip link set can1 up type can bitrate 1000000
```

Use the cansend command to send data:

```
cansend can1 123#DEADBEEF12345678
```

Explanation:

- (1) can0 device
- (2) 123: Frame ID
- (3) DEADBEEF12345678: Frame data

序号	时间标识	源通道	帧ID	CAN类型	方向	长度	数据
				全部	全部		
0	0.000000	0	0x123	CAN	Rx	8	DE AD BE EF 12 34 56 78
1	1.847600	0	0x123	CAN	Rx	8	DE AD BE EF 12 34 56 78
2	3.192800	0	0x123	CAN	Rx	8	DE AD BE EF 12 34 56 78
3	4.411900	0	0x123	CAN	上位机接收CAN数据	8	DE AD BE EF 12 34 56 78
4	5.668500	0	0x123	CAN	Rx	8	DE AD BE EF 12 34 56 78
5	6.843200	0	0x123	CAN	Rx	8	DE AD BE EF 12 34 56 78
6	8.393700	0	0x123	CAN	Rx	8	DE AD BE EF 12 34 56 78


```

root@atk-dlam62l:~# ip link set can1 up type can bitrate 1000000
root@atk-dlam62l:~# cansend can1 123#DEADBEEF12345678
root@atk-dlam62l:~# cansend can1 123#DEADBEEF12345678
root@atk-dlam62l:~# cansend can1 123#DEADBEEF12345678
root@atk-dlam62l:~# cansend can1 123#DEADBEEF12345678
root@atk-dlam62l:~# cansend can1 123#DEADBEEF12345678
root@atk-dlam62l:~# cansend can1 123#DEADBEEF12345678 开发板发送CAN数据
root@atk-dlam62l:~# cansend can1 123#DEADBEEF12345678
root@atk-dlam62l:~#

```

Figure 4.7-2 Development Board CAN Data Transmission Test

For data reception, the candump command can be used to receive data.

```
candump -ta can1
```

Explanation:

(1) -ta: The letter **t** indicates printing time, and **a** enables ASCII output.

The experimental results, obtained by setting the CAN rate to 1,000,000 bps using the ZLG USBCANFD 200U CAN analyzer, are shown in the figure below.

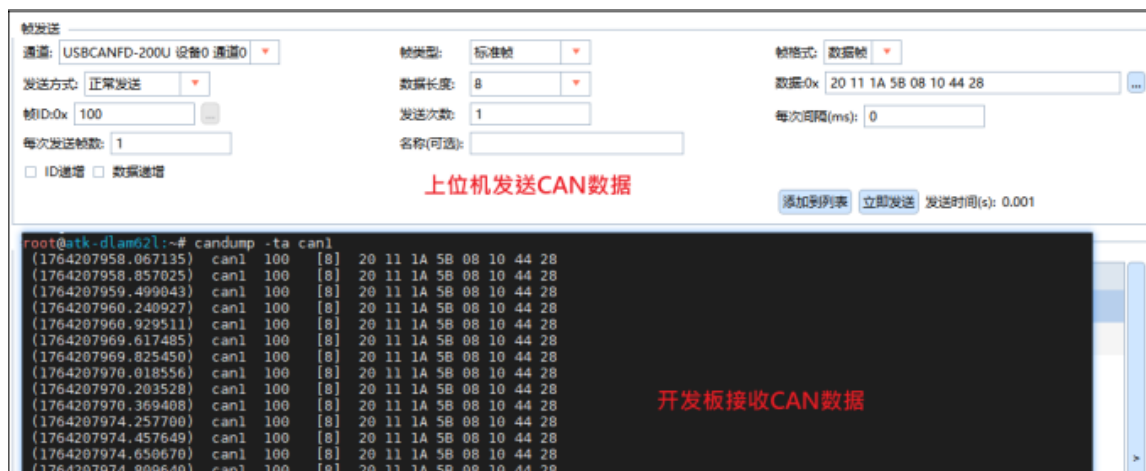


Figure 4.7-3 Development Board CAN Data Reception Test

4.7.2 CAN-FD Test

CAN has already been enabled in the previous section. If you do not want to reboot the development board, you can use the following commands to disable CAN first.

```
ifconfig can1 down
```

Configure the CAN FD rate; test other rates as needed.

```
ip link set can1 up type can bitrate 1000000 dbitrate 5000000 fd on
```

Set the baud rate to 1000 kBit/s and the data baud rate to 5000 kBit/s for testing.

```
root@atk-dlam62l:~#
root@atk-dlam62l:~# ip link set can1 up type can bitrate 1000000 dbitrate 5000000 fd on
root@atk-dlam62l:~#
```

Figure 4.7-4 Enable and configure the CAN baud rate

For CAN FD data transmission, please copy the following commands

```
cansend can1 123##300.11.22.33.44.55.66.77.88.99.aa.bb.cc.dd.ee.ff.00.11.22.33.44.55.66.77.88.99.aa.bb.cc.dd.ee.ff.00.11.22.33.44.55.66.77.88.99.aa.bb.cc.dd.ee.ff
```

Explanation:

- (1) can0: CAN device
- (2) 123: Frame ID
- (3) 3: Flags
- (4) 00.11.22....: Frame data

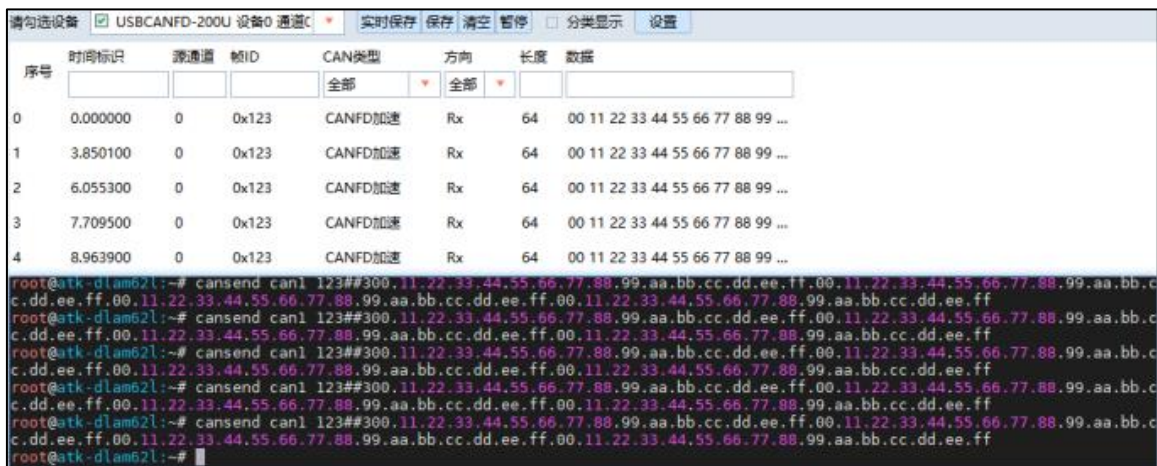


Figure 4.7-5 Development Board CAN-FD Data Transmission Test

For data reception, the `candump` command can be used to receive data.

`candump -ta can1`

Explanation:

(1) `-ta`: **t** stands for printing time, and **a** stands for enabling ASCII output.

The following are the test results using the ZLG USBCANFD 200U CAN analyzer with a maximum baud rate of 1000 kBit/s and a maximum data baud rate of 5000 kBit/s.



Figure 4.7-6 Development Board CAN-FD Data Reception Test

4.8 RTC clock testing

ATK-DLAM62L development board has RTC clock chip PCF8563, which belongs to the external RTC clock of the chip. If you need to improve the accuracy of the clock, you need to use high precision crystal.

Please check whether the RTC button battery is installed on the development board bottom plate. You can also use a multimeter to check whether the RTC battery has power, and it is normal to measure about 3.3v. Prevent the RTC battery power can not save time. (Note that the battery is a consumable, if no power, please replace the button battery)

The Linux system is divided into two clocks, one is the system time (software clock), the other is the hardware clock (hardware clock). The `date` and `hwclock` commands can be used to view and set the

system and hardware time, respectively. The system clock will disappear when the power is off, and the RTC clock will run for a long time when there is a battery. The system clock is synchronized with the RTC clock when the system is restarted.

To check the system clock, use the date command. (As long as the development board can access the Internet, the process of the system will automatically update the time)

```
date
```

```
root@atk-dlam62l:~# date
Wed Nov 26 17:03:28 CST 2025
root@atk-dlam62l:~#
```

Figure 4.8-1 Check the system time

If there is no automatic update time (if the board has an Internet connection), you can run the following command:

```
ntpdate -u cn.pool.ntp.org
```

```
root@atk-dlam62l:~# ntpdate -u cn.pool.ntp.org
26 Nov 17:04:21 ntpdate[1116]: adjust time server 84.16.73.33 offset -0.004184 sec
root@atk-dlam62l:~#
```

Figure 4.8-2 Update time

You can set the current system time by running the following command without a cable.

```
date -s "2025-11-26 17:04:00"
```

Set RTC hardware time (write system time into hardware time)

```
hwclock -w
```

Look at the hardware clock to see if it is the one set by hwclock-w above.

```
hwclock
```

```
root@atk-dlam62l:~# date -s "2025-11-26 17:04:00"
Wed Nov 26 17:04:00 CST 2025
root@atk-dlam62l:~#
root@atk-dlam62l:~#
root@atk-dlam62l:~# hwclock -w
root@atk-dlam62l:~# hwclock
Wed Nov 26 17:04:07 2025 0.000000 seconds
```

Figure 4.8-3 Hardware time

Note that the time zone of system time and RTC time is UTC. China time is 8 hours faster than UTC, so the program needs to add 8 hours.

4.9 Audio Testing

4.9.1 Playing the test

Audio test script is pre-installed in the factory system, which is used to quickly run audio test in the terminal to verify the working state of audio equipment. To run the test script, you need to go through a series of steps, first navigate to the specific script directory (/opt/shell/audio), and then execute the script file. The command will look as follows:

```
cd /opt/shell/
./alientek_audio.sh
```

```

root@atk-dlam62l:/opt/shell/audio# ./atk_audioTest.sh
=====
1 -> Line 1 耳机录音5秒后自动播放
2 -> Line 2 MIC录音5秒后自动播放
3 -> 播放测试音乐 SoaringFree.mp3
其他键 或 Ctrl+C -> 退出
=====
请选择 (1/2/3) : █

```

Figure 4.9-1 alientek audio.sh script

Enter 3 to start playing the test.

```

root@ATK-DLT113IS:/opt/shell# ./alientek_audio.sh
请选择要运行的测试:
1. 播放测试
2. MIC1录音测试
3. MIC2录音测试
4. 耳机录音测试
5. FMIN录音测试
6. LINEIN录音测试
请输入你的选择 (1-6): 1
运行播放测试...
Press 'k' to see a list of keyboard shortcuts.
Now playing /opt/ui/src/apps/resource/audio/陈楚生 - 有没有人告诉你.mp3
[ 6930.060871] [SNDCodec][sunxi_card_hw_params][635]:stream_flag: 0
Redistribute latency...
[ 6931.466876] [SNDCodec][sunxi_check_hs_detect_status][192]:plugin --> switch:3
[ 6932.866832] [SNDCodec][sunxi_check_hs_detect_status][230]:plugin --> switch:0
0:00:04.7 / 0:05:44.0

```

Figure 4.9-2 Play test

4.9.2 Recording test

Recording is the same as playing, and you can use a script to record. First, navigate to the specific script directory (/opt/shell/audio), and the command is as follows:

```

cd /opt/shell/
./alientek_audio.sh

```

```

root@ATK-DLT113IS:/opt/shell# ./alientek_audio.sh
请选择要运行的测试:
1. 播放测试
2. MIC1录音测试
3. MIC2录音测试
4. 耳机录音测试
5. FMIN录音测试
6. LINEIN录音测试
请输入你的选择 (1-6): █

```

Figure 4.9-3 Choose the appropriate configuration

Enter 1 to start the on-board headphone recording test, and enter 2 to start the on-board MIC recording test. After selection, recording will start for 5 seconds, and the recorded audio will be played automatically after recording is completed.

```

root@atk-dlam62l:/opt/shell/audio# ./atk_audioTest.sh
=====
1 -> Line 1 耳机录音5秒后自动播放
2 -> Line 2 MIC录音5秒后自动播放
3 -> 播放测试音乐 SoaringFree.mp3
其他键 或 Ctrl+C -> 退出
=====
请选择 (1/2/3) : 2
已切换到 Line 2 (Line 2P + Line 2N)
开始录音 5 秒 (Line2) ...
录音完成! 正在播放...
Line2 测试完成
请选择 (1/2/3) : █

```

Figure 4.9-4 Recording test

4.10 Onboard wifi test

First, use the following command to enable the WiFi node. If the error `ifconfig: SIOCSIFFLAGS: Operation not possible due to RF-kill` occurs, you need to disable the radio frequency first.

```

ifconfig wlan0 up
rfkill unblock all

```

4.10.1 Station mode

Scan for Wi-Fi hotspots. The `iw` command provided by the file system can be used to scan the hotspot names.

To view complete hotspot information, first enable `wlan0` and use the `iw wlan0 scan` command. Here, `grep` is used to filter out some information.

If the error "ifconfig: SIOCSIFFLAGS: Operation not possible due to RF-kill" occurs, you need to disable the radio frequency first.

```
rfkill unblock all // Disable radio frequency (only execute if the above error occurs).
```

```
ifconfig wlan0 up
```

```
iw wlan0 scan | grep SSID
```

The following figure scans the hot spot name "ALIENTEK-linux", where "x00\x00\..." Because the Chinese characters can not be displayed. So the user's hot spots had better be composed of English characters.

```
root@atk-dlam62l:~# iw wlan0 scan | grep SSID
SSID: 
SSID: alientek-linux
SSID: ALIENTEK-XXXX
SSID: 
SSID: 
SSID: 
SSID: * U.S. ...
SSID: 
SSID: 
SSID: 
SSID: 
SSID: ALI
SSID: 
SSID:
```

Figure 4.10-1 Scanning hot spots

Edit the hotspot information to connect to, change `/etc/wpa_supplicant.conf` to the following, and type `:wq` save and exit. Please change the ssid and psk of personal hotspot.

Explanation:

- ssid is the wireless network name
- psk is the wireless network password

```
vi /etc/wpa supplicant.conf
```

Enter the following:

```
ctrl interface=/var/run/wpa supplicant
```

ap scan=1

```
network={
    ssid="ALIENTEK-linux"
    psk="1590202****"
}
```

If the hotspot is passwordless, change `/etc/wpa_supplicant.conf` to the following

```
ctrl interface=/var/run/wpa supplicant
```

ap scan=1

```
network={
    ssid="ALIENTEK-linux"
```

```
key_mgmt=NONE
```

Execute the following commands to connect to the hotspot and get the ip address.

```
wpa_supplicant -Dnl80211 -c /etc/wpa_supplicant.conf -i wlan0 &
udhcpc -i wlan0
```

```
wlan0: CTRL-EVENT-STARTED-CHANNEL-SWITCH freq=5745 ht_enabled=1 ch_offset=1 ch_width=40 MHz
wlan0: Associated with 22:cd:d2:b7:6e:10
wlan0: CTRL-EVENT-SUBNET-STATUS-UPDATE status=0
wlan0: WPA: Key negotiation completed with 22:cd:d2:b7:6e:10 [PTK=CCMP GTK=CCMP]
wlan0: CTRL-EVENT-CONNECTED - Connection to 22:cd:d2:b7:6e:10 completed [id=0 id_str=]

root@atk-dlam62l:~#
root@atk-dlam62l:~#
root@atk-dlam62l:~# udhcpc -i wlan0
udhcpc: started, v1.36.1
udhcpc: broadcasting discover
udhcpc: broadcasting select for 192.168.3.101, server 192.168.3.1
udhcpc: lease of 192.168.3.101 obtained from 192.168.3.1, lease time 86400
deleting routers
adding dns 192.168.3.1
root@atk-dlam62l:~#
```

Figure 4.10-2 Connect to a WIFI hotspot

If you see the above result, you have connected to the WIFI hotspot and obtained the ip information. We can also use the ifconfig directive to see the ip address obtained by our wlan0.

```
wlan0    Link encap:Ethernet  HWaddr 4C:A3:8F:7B:40:B0
          inet addr:192.168.3.101  Bcast:192.168.3.255  Mask:255.255.255.0
          inet6 addr: fe80::4ea3:8fff:fe7b:40b0/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:54 errors:0 dropped:0 overruns:0 frame:0
          TX packets:16 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:6324 (6.1 KiB)  TX bytes:2304 (2.2 KiB)

wlan1    Link encap:Ethernet  HWaddr 4E:A3:8F:7B:40:B0
          UP BROADCAST MULTICAST  MTU:1500  Metric:1
          RX packets:0 errors:0 dropped:0 overruns:0 frame:0
          TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)
```

Figure 4.10-3 Retrieve the ip address

ping Baidu to test connectivity, or ping gateway to test WIFI connectivity. Note: If the ping fails to Baidu, please restart the development board, do not plug in the network cable, and reconnect to the WIFI! By default, only one network device is allowed to connect to the public network.

```
ping www.baidu.com -I wlan0
```



```

root@atk-dlam62l:~# ping www.baidu.com -I wlan0
PING www.baidu.com (183.2.172.17): 56 data bytes
64 bytes from 183.2.172.17: seq=0 ttl=51 time=9.999 ms
64 bytes from 183.2.172.17: seq=1 ttl=51 time=9.831 ms
64 bytes from 183.2.172.17: seq=2 ttl=51 time=12.774 ms
64 bytes from 183.2.172.17: seq=3 ttl=51 time=10.535 ms
64 bytes from 183.2.172.17: seq=4 ttl=51 time=12.489 ms
64 bytes from 183.2.172.17: seq=5 ttl=51 time=11.788 ms
64 bytes from 183.2.172.17: seq=6 ttl=51 time=9.294 ms
64 bytes from 183.2.172.17: seq=7 ttl=51 time=9.501 ms
^C
--- www.baidu.com ping statistics ---
8 packets transmitted, 8 packets received, 0% packet loss
round-trip min/avg/max = 9.294/10.776/12.774 ms

```

Figure 4.10-4 ping Baidu test

4.10.2 Hotspot (SoftAP) mode

SoftAP mode for WiFi is an operating mode that allows a WiFi device to function like a wireless router. In SoftAP mode, the WiFi device can act as a hotspot, enabling other devices to connect and use the network services it provides. Generally, SoftAP mode is suitable for scenarios where a device needs to be converted into a wireless hotspot, such as smart home devices, wireless printers, game consoles, and more. In SoftAP mode, the device can be configured with parameters including SSID and password, allowing other devices to connect and perform data transmission and communication through the network services it offers.

Since SoftAP mode enables a device to operate like a wireless router, it is commonly used in specialized scenarios such as network debugging and data collection.

The hotspot-enabling scripts developed by ALIENTEK are all stored in the `/opt/shell/wifi` directory, as shown in the figure below.

```

root@atk-dlam62l:/opt/shell/wifi# ls
dnsmasq.conf  hostapd.conf  wifi_apmode.sh
root@atk-dlam62l:/opt/shell/wifi#

```

Figure 4.10-5 WiFi Script

The content of the **dnsmasq.conf** file is as follows:

```

interface=wlan0
dhcp-range=192.168.4.2,192.168.4.254,255.255.255.0,24h
port=49153

```

The content of the **hostapd.conf** file is as follows:

```

interface=wlan0 ssid=alientek_softap driver=nl80211 channel=6
hw_mode=g ignore_broadcast_ssid=0 auth_algs=1
wpa=3 wpa_passphrase=12345678
wpa_key_mgmt=WPA-PSK
wpa_pairwise=TKIP
rsn_pairwise=CCMP

```

The content of the **wifi_apmode.sh** file is as follows:

```

echo 0 > /sys/class/rfkill/rfkill1/state
sleep 2
echo 1 > /sys/class/rfkill/rfkill1/state

```




Figure 4.10-7 Mobile connection hotspot

As you can see, the board's IP address is **192.168.4.1** for wlan0, which is set by the `wifi_apmode.sh` script, acting as the gateway (i.e., router). Meanwhile, the phone has been assigned an IP address of **192.168.4.4** by the development board.

Note that although this hotspot can be connected, it is **not for accessing the Internet** — this demo is limited to connections within the local area network only.

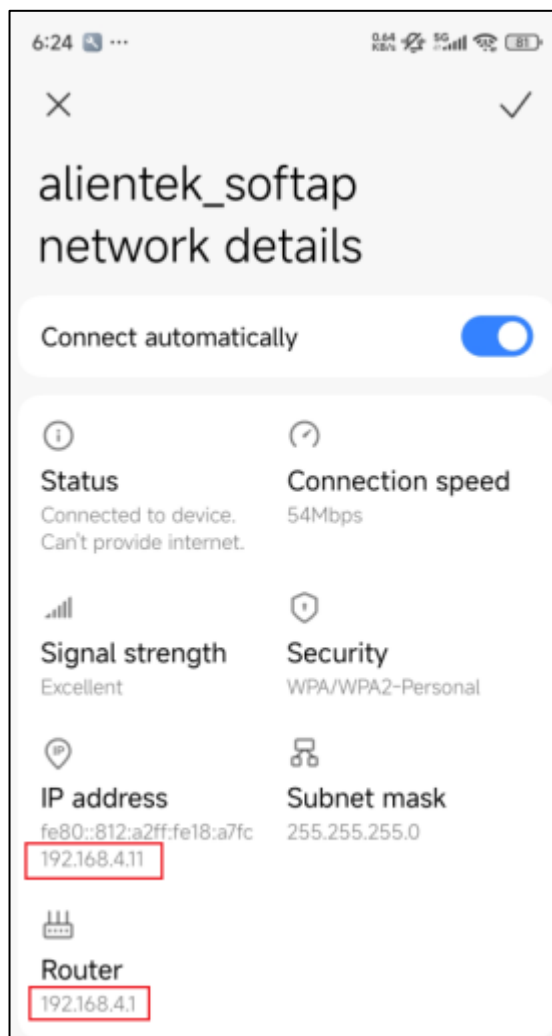


Figure 4.10-8 WiFi Details

Now that we have enabled AP mode, how do we disable it? In the previous scripts, we used **hostapd** and **dnsmasq** to start AP mode as background processes. Therefore, we need to stop these two processes to disable AP mode. Run the following commands to terminate the processes.

```
killall hostapd dnsmasq
```

```
root@atk-dlam62l:/opt/shell/wifi#
root@atk-dlam62l:/opt/shell/wifi# killall hostapd dnsmasq
root@atk-dlam62l:/opt/shell/wifi#
```

Figure 4.10-9 Disable AP Mode

You can also directly use the **iw** command to set WiFi to client mode. However, it is better to shut down the two processes mentioned above to prevent resource occupation!

```
iw dev wlan0 set type managed
```

```
root@atk-dlam62l:/opt/shell/wifi#
root@atk-dlam62l:/opt/shell/wifi# iw dev wlan0 set type managed
root@atk-dlam62l:/opt/shell/wifi#
```

Figure 4.10-10 Set Client Mode

At this point, we use the **iw dev** command to view the wireless network interface. If the type of the wireless network is shown as **managed**, it means client mode, which is also the **station mode** mentioned above.

iw dev

```

root@atk-dlam62l:/opt/shell/wifi# iw dev
phy#0
    Interface wlan1
        ifindex 7
        wdev 0x2
        addr 4e:a3:8f:7b:40:b0
        type managed
        txpower -100.00 dBm
    Interface wlan0
        ifindex 6
        wdev 0x1
        addr 4c:a3:8f:7b:40:b0
        type managed
        txpower -100.00 dBm

```

Figure 4.10-11 iw dev

4.11 Onboard Bluetooth test

You can see from the boot print that the Bluetooth driver is loaded.

```

[ 5.980052] Bluetooth: HCI device and connection manager initialized
[ 5.987246] Bluetooth: HCI socket layer initialized
[ 5.992259] Bluetooth: L2CAP socket layer initialized
[ 5.997402] Bluetooth: SCO socket layer initialized
[ 6.045077] rtk_btusb: btusb_probe intf->cur_altsetting->desc.bInterfaceNumber 0
[ 6.053636] rtk_btusb: btusb_open hdev->promisc ==0
[ 6.058611] rtk_btusb: chip type value: 0x76
[ 6.064380] usbcore: registered new interface driver rtk_btusb
[ 6.082540] rtk_btusb: config filename rtl8733bu_config
[ 6.091314] Loaded X.509 cert 'sforshee: 00b28ddf47aef9cea7'
[ 6.092526] rtk_btusb: no bdaddr file /opt/bdaddr
[ 6.097920] Loaded X.509 cert 'wens: 61c038651aabdcf94bd0ac7ff06c7248db18c600'
[ 6.102162] rtk_btusb: Origin cfg len 14
[ 6.109542] platform regulatory.0: Direct firmware load for regulatory.db failed with error -2
[ 6.121758] rtk_btusb: New cfg len 14
[ 6.122097] cfg80211: failed to load regulatory.db
[ 6.137297] rtk_btusb: This is not 8723a, use new patch style!
[ 6.406885] usbcore: registered new interface driver rtl8733bu
[ 6.429559] rtk_btusb: chip type value: 0x76
[ 6.433931] rtk_btusb: btusb_open set HCI UP RUNNING
done[ 6.439089] rtk_btcoex: Open BTCOE

```

Figure 4.11-1 Bluetooth initialization information

To see if Bluetooth is enabled, run the following code:

```
hciconfig -a
```

```

root@atk-dlam62l:~# hciconfig -a
hci0: Type: Primary Bus: USB
      BD Address: 4C:A3:8F:7B:C0:B1 ACL MTU: 1021:6 SCO MTU: 255:12
      UP RUNNING
      RX bytes:818 acl:0 sco:0 events:62 errors:0
      TX bytes:2283 acl:0 sco:0 commands:62 errors:0
      Features: 0xff 0xff 0xff 0xfa 0xdb 0xbf 0x7b 0x87
      Packet type: DM1 DM3 DM5 DH1 DH3 DH5 HV1 HV2 HV3
      Link policy: RSWITCH HOLD SNIFF PARK
      Link mode: PERIPHERAL ACCEPT
      Name: 'BlueZ 5.79'
      Class: 0x000000
      Service Classes: Unspecified
      Device Class: Miscellaneous,
      HCI Version: 5.2 (0xb) Revision: 0xd8c7
      LMP Version: 5.2 (0xb) Subversion: 0x6fec
      Manufacturer: Realtek Semiconductor Corporation (93)

```

Figure 4.11-2 View Bluetooth capabilities

4.11.1 Bluetooth pairing

Bluetooth pairing requires the bluetoothd service, which runs in the background as a system service and is responsible for managing the Bluetooth adapter and handling the connection, pairing, and communication of Bluetooth devices. It is one of the core components related to Bluetooth in Linux system. bluetoothd is stored in /usr/libexec/bluetoothd, and we can start it in the background by running the following command:

```
/usr/libexec/bluetooth/bluetoothd -C &
```

Enter the following command to enter the bluetoothctl interactive mode.

```
bluetoothctl
```

Execute the following commands in turn:

```
power on           # Turn on the power
agent on           # Enable proxies
discoverable on    # The opening can be detected
scan on            # Enable scanning device
```

```
root@atk-dlam62l:~# bluetoothctl
[bluetooth]# hcib0 new_settings: powered bondable ssp br/edr le secure-conn wide-band-speech cis-central cis-peripheral
[bluetooth]# Agent registered
[bluetooth]# [CHG] Controller 4C:A3:8F:7B:C0:B1 Pairable: yes
[bluetooth]# power
Missing on/off argument
[bluetooth]# agent on
Agent is already registered
[bluetooth]# discoverable on
[bluetooth]# hcib0 new_settings: powered connectable bondable ssp br/edr le secure-conn wide-band-speech cis-central cis-peripheral
[bluetooth]# hcib0 new_settings: powered connectable discoverable bondable ssp br/edr le secure-conn wide-band-speech cis-central cis-peripheral
[bluetooth]# Changing discoverable on succeeded
[bluetooth]# [CHG] Controller 4C:A3:8F:7B:C0:B1 Discoverable: yes
[bluetooth]# scan on
[bluetooth]# SetDiscoveryFilter success
[bluetooth]# [ 1571.348782] rtk_btcoex: hci (periodic)inq start
hci0 type 7 discovering on
[bluetooth]# Discovery started
[bluetooth]# [CHG] Controller 4C:A3:8F:7B:C0:B1 Discovering: yes
[bluetooth]# [NEW] Device E1:10:4A:5E:74:21-10-4A
[bluetooth]# [NEW] Device 4C:A3:8F:7B:C0:B1 NPgLEzUePdME1coKU40M6LBI
[bluetooth]# [NEW] Device CD:29:9A:93:41-A2
[bluetooth]# [CHG] Device E1:10:4A:5E:74:21 TxPower: 0x0000 (0)
[bluetooth]# [CHG] Device EC:10:4A:5E:74:21 A Name: ATK-BLE02
```

Figure 4.11-3 bluetoothctl interaction mode

Wait until you scan your Bluetooth device and enter scan off to stop scanning.

```
scan off
```

Enter 'devices' to print out the scanned Bluetooth devices.

As shown in the figure below, I have scanned the Bluetooth MAC address of my own Android phone. After confirmation, pairing can be performed.

Different mobile phone systems may have different interception or data protection policies for Bluetooth (such as the iOS system). Please ensure that the phone's Bluetooth is not blocked.

```
Device 4C:A3:8F:7B:C0:B1 EDIFIER BLE
Device 4C:A3:8F:7B:C0:B1 蓝牙测试
Device 48:D5:16:10:8B NPgLEzUePdME1coKU40M6LBI
Device 4C:A3:8F:7B:C0:B1 47-EA-27-F7-D8-BB
```

Figure 4.11-4 Discovered Bluetooth Device

Your Bluetooth device has been found; pairing can be performed after confirmation.

Enter the following command to pair.

```
pair 68:C4:XX:XX:XX:XX # Fill in your Bluetooth MAC
```

```
[bluetooth]# pair 68:C4:4C:E8:0E:46
Attempting to pair with 68:C4:4C[ 241.641473] rtk_btusb: btusb_notify: hci0 evt 1
:E8:0E:46
[ 241.650739] rtk_btcoex: hci create connection, start paging
[bluetooth]# [ 244.580666] rtk_btcoex: connected, handle 0001, status 0x00
[ 244.587169] rtk_btcoex: Page success
[ 244.650677] rtk_btcoex: io capability request
[CHG] Device 68:C4:4C:E8:0E:46 Connected: yes
Request confirmation
[agent] Confirm passkey 230916 (yes/no): yes
[蓝牙测试]# [ 256.063647] rtk_btcoex: link key notify
[ 256.095619] rtk_btcoex: l2cap op 2, len 16, out 0
[ 256.101063] rtk_btcoex: RX l2cap conn req, hndl 0x0001, PSM 0x0001, scid 0x0043
[ 256.101167] rtk_btcoex: l2cap op 3, len 20, out 1
```

Figure 4.11-5 Bluetooth pairing

The phone also pops up the pairing code.

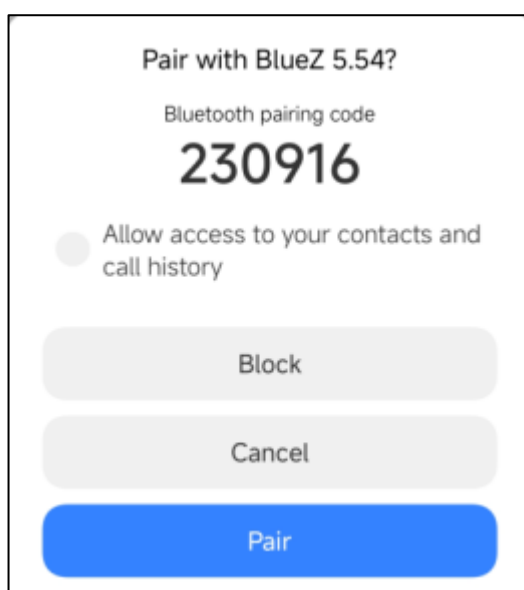


Figure 4.11-6 Cell phone pairing code

After a successful match, we can manually connect the development board's Bluetooth from the mobile phone.

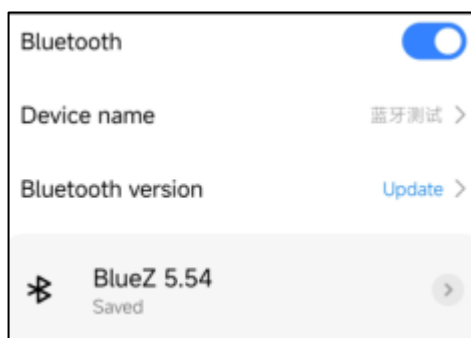


Figure 4.11-7 Connect the development board's Bluetooth on the mobile phone

Once connected to your Bluetooth device, enter scan off to stop scanning, then enter exit to exit the interactive mode.

scan off

exit

```
[蓝牙测试]# [CHG] Device 68:C4:4C:E8:0E:46 ServicesResolved: no
[bluetooth]# [CHG] Device 68:C4:4C:E8:0E:46 Connected: no
[bluetooth]# exit
root@atk-dlam62l:~#
```

Figure 4.11-8 Exit the Bluetooth tool

Test the Bluetooth connection using the **l2ping** utility. This tool requires the target Bluetooth device to support the L2CAP echo function and have relevant Bluetooth permissions enabled. The test should be performed at a moderate distance (usually within 10 meters), without obvious physical obstacles or electromagnetic interference sources. The target device used here is an Android phone.

l2ping 68:C4:4C:E8:0E:46 # Please enter your own Bluetooth MAC address.

```
root@atk-dlam62l:~# l2ping 68:C4:4C:E8:0E:46
[ 1894.575716] rtk_btcoex: hci create connection, start paging
[ 1897.514532] rtk_btcoex: connected, handle 0001, status 0x00
[ 1897.520209] rtk_btcoex: Page success
Ping: 68:C4:4C:E8:0E:46 from 4C:A3:8F:7B:C0:B1 (data size 44) ...
44 bytes from 68:C4:4C:E8:0E:46 id 0 time 21.16ms
44 bytes from 68:C4:4C:E8:0E:46 id 1 time 15.94ms
44 bytes from 68:C4:4C:E8:0E:46 id 2 time 13.55ms
44 bytes from 68:C4:4C:E8:0E:46 id 3 time 11.13ms
44 bytes from 68:C4:4C:E8:0E:46 id 4 time 23.78ms
44 bytes from 68:C4:4C:E8:0E:46 id 5 time 20.74ms
44 bytes from 68:C4:4C:E8:0E:46 id 6 time 18.78ms
44 bytes from 68:C4:4C:E8:0E:46 id 7 time 19.91ms
^C8 sent, 8 received, 0% loss
```

Figure 4.11-9 Bluetooth l2ping Test

If the Bluetooth connection is ineffective or fails, clear the Bluetooth information in `bluetoothctl`, then run `scan` on again to reconnect.

devices	#View Paired Bluetooth Devices
disconnect E0:40:07:B0:07:32	# Command used to disconnect from a specific Bluetooth device
untrust E0:40:07:B0:07:32	# Remove a Bluetooth device from the trusted list
remove E0:40:07:B0:07:32	# Remove a Bluetooth device from the pairing list.

4.12 4G module EC20 test

The Quectel EC20 4G module is a high performance, high reliability wireless communication module designed for embedded systems. It supports 4G LTE network, has excellent data transmission speed and stability, and can be widely used in the Internet of things, Internet of vehicles, industrial automation and other fields.

LED flashing corresponding to LED_WWAN by default:

Pin working state	The indicated network status
Slow flash (200ms high /1800ms low)	Finding network status
Slow flash (1800ms high /200ms low)	Standby mode
Flash (125ms high /125ms low)	Data transfer mode
High level	On the phone

Preparation before the experiment: EC20 4G module, 4G network card and antenna (for amplifying signal).

4G module interface is reserved on the base board of ATK-DLT113IS. ME3630-W, EC20 and other 4G modules are installed. Prepare the EC20 module, pay attention to the need to buy the antenna,

the module alone will not work properly! (Note: there are many types of EC20 modules, EC20-CE modules have been tested at present, including EC20-CE series and a variety of modules, different module functions are not the same, such as the support of the operator is not the same, please consult the seller for details), other EC20 series, please test by yourself, theoretically driven the same, there is a need to find far technical support.) . Insert the EC20 4G module into the 4G module interface and screw on the screw. Ensure that the 4G module is connected to the seat interface. Please use the original antenna and connect the antenna to the MAIN interface of the 4G module.

- Correctly insert the 4G card and plug in the module, the WWAN LED on the bottom board will light green after the development board is started. If the green light of the WWAN LED does not light up, please check whether the module is correctly connected and inserted, whether the 4G card is inserted, and whether the antenna is connected properly. The development board must be connected with the 12V power supply provided with it, and cannot only use the serial port USB_TTL for power supply.
- Before the 4G module test, insert the 4G card into the SIM card slot of the bottom board, and then plug in the EC20 4G module. At the same time, plug in the antenna, and the antenna is connected to the MAIN of the module. After inserting the 4G card and antenna correctly, the WWAN LED on the bottom board will light up green light after the development board is started. If the light does not light up, please check whether the 4G card is inserted in the right position and whether the antenna is connected correctly, and then try to reinsert the module. Must plug in the development board used power! Otherwise, the power supply is insufficient and the module cannot work properly. The module installation looks like the following:

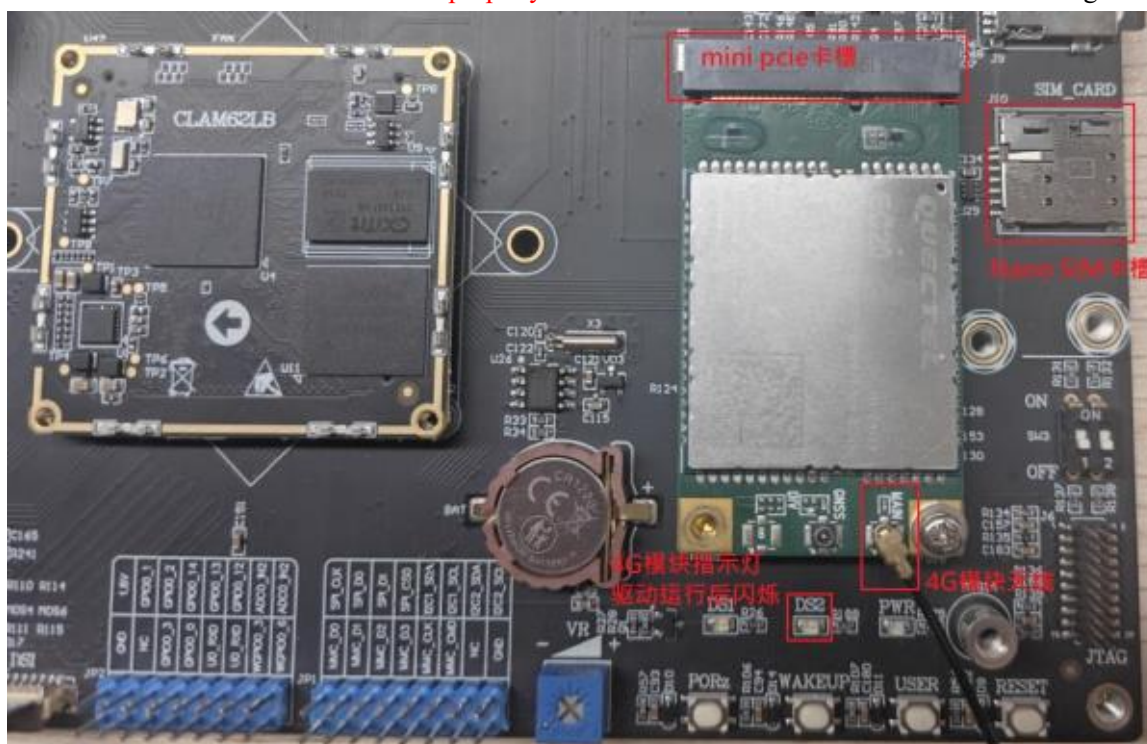


Figure 4.12-1 EC20 installation diagram

We can print information on the serial port and see the following print information:


```

5.488336] usbcore: registered new interface driver option
5.494420] usbserial: USB Serial support registered for GSM modem (1-port)
5.501815] option 1-1.4:1.0: GSM modem (1-port) converter detected
5.508811] usb 1-1.4: GSM modem (1-port) converter now attached to ttyUSB0
5.516169] option 1-1.4:1.1: GSM modem (1-port) converter detected
5.523118] usb 1-1.4: GSM modem (1-port) converter now attached to ttyUSB1
5.530432] option 1-1.4:1.2: GSM modem (1-port) converter detected
5.537354] usb 1-1.4: GSM modem (1-port) converter now attached to ttyUSB2
5.544689] option 1-1.4:1.3: GSM modem (1-port) converter detected
5.552290] rtk_btusb: btusb_probe intf->cur_altsetting->desc.bInterfaceNumber 0
5.552621] usb 1-1.4: GSM modem (1-port) converter now attached to ttyUSB3
5.570092] usbcore: registered new interface driver rtk_btusb
5.580859] rtk_btusb: btusb_open hdev->promisc ==0
5.585872] rtk_btusb: chip type value: 0x76

```

Figure 4.12-2 USB information printed by EC20

Then check to see if /dev/ttyusb0-3 nodes are generated

```
ls /dev/ttyUSB*
```

```

root@atk-dlam62l:~# ls /dev/ttyUSB*
/dev/ttyUSB0 /dev/ttyUSB1 /dev/ttyUSB2 /dev/ttyUSB3
root@atk-dlam62l:~#

```

Figure 4.12-3 Look at the USB generated node

The functions of the four ttyUSB are shown in the following figure, not all these functions are tested, here we only test the Internet function. Please refer to the EC20 4G module manual for details.

Product			USB Driver	Interface
UC15	VID: 0x05c6	PID: 0x9090	USB Serial	ttyUSB0 → DM
UC20	VID: 0x05c6	PID: 0x9003		
EC25	VID: 0x2c7c	PID: 0x0125		ttyUSB1 → For GPS NMEA message output
EC21	VID: 0x2c7c	PID: 0x0121		
EC20	VID: 0x05c6	PID: 0x9215		ttyUSB2 → For AT commands
EC20 R2.0	VID: 0x2c7c	PID: 0x0125		
EG91	VID: 0x2c7c	PID: 0x0191		ttyUSB3 → For PPP connections or AT commands
EG95	VID: 0x2c7c	PID: 0x0195		
UC20	VID: 0x05c6	PID: 0x9003	GobiNet or QMI WWAN	
EC25	VID: 0x2c7c	PID: 0x0125		
EC21	VID: 0x2c7c	PID: 0x0121		
EC20	VID: 0x05c6	PID: 0x9215		
EC20 R2.0	VID: 0x2c7c	PID: 0x0125		
EG91	VID: 0x2c7c	PID: 0x0191		
EG95	VID: 0x2c7c	PID: 0x0195		
EG06	VID: 0x2c7c	PID: 0x0306		
EP06	VID: 0x2c7c	PID: 0x0306		
EM06	VID: 0x2c7c	PID: 0x0306		
BG96	VID: 0x2c7c	PID: 0x0296		

Figure 4.12-4 Functional diagram of four-way ttyUSB

4.12.1 ppp dial-up Internet access

Enter the directory "/opt/shell/4G", which stores the 4G module script. If you do not see the 4G directory, please update the factory system by yourself.

```
cd /opt/shell/4G
```

The ppp dials are mainly ppp-on-1000, ppp-on-10010, and ppp-on-10086. The three scripts are configured with different APN values for different operators. ppp-on-1000, ppp-on-10010 and ppp-on-

10086 are the scripts that need to be executed by the telecom card, the script that needs to be executed by the Unicom card and the script that needs to be executed by the mobile card, respectively.

For example, if a mobile card is used in this test, the script executed would be ppp-on-10000.

./ppp-on-10000 &

```
Connect: ppp0 <-> /dev/ttyUSB2
sent [LCP ConfReq id=0x1 <asyncmap 0x0> <magic 0xe5647ad3> <pcomp> <accomp>]
rcvd [LCP ConfReq id=0x0 <asyncmap 0x0> <auth chap MD5> <magic 0x3338c26> <pcomp> <accomp>]
sent [LCP ConfAck id=0x0 <asyncmap 0x0> <auth chap MD5> <magic 0x3338c26> <pcomp> <accomp>]
rcvd [LCP ConfAck id=0x1 <asyncmap 0x0> <magic 0xe5647ad3> <pcomp> <accomp>]
rcvd [LCP DiscReq id=0x1 magic=0x3338c26]
rcvd [CHAP Challenge id=0x1 <99b0397d5cbbcb7da10f9f19872c53e4>, name = "UMTS_CHAP_SRVR"]
sent [CHAP Response id=0x1 <23bb09cc8c889669ec59c548be3df89b>, name = "Anyname"]
rcvd [CHAP Success id=0x1 ""]
CHAP authentication succeeded
CHAP authentication succeeded
sent [IPCP ConfReq id=0x1 <addr 0.0.0.0> <ms-dns1 0.0.0.0> <ms-dns2 0.0.0.0>]
sent [IPV6CP ConfReq id=0x1 <addr fe80::fc49:896b:7e79:5ad8>]
rcvd [IPCP ConfReq id=0x0]
sent [IPCP ConfNak id=0x0 <addr 0.0.0.0>]
rcvd [IPCP ConfNak id=0x1 <addr 10.6.237.231> <ms-dns1 202.96.128.86> <ms-dns2 202.96.134.133>]
sent [IPCP ConfReq id=0x2 <addr 10.6.237.231> <ms-dns1 202.96.128.86> <ms-dns2 202.96.134.133>]
rcvd [IPCP ConfReq id=0x1]
sent [IPCP ConfAck id=0x1]
rcvd [IPCP ConfAck id=0x2 <addr 10.6.237.231> <ms-dns1 202.96.128.86> <ms-dns2 202.96.134.133>]
Could not determine remote IP address: defaulting to 10.64.64.64
local IP address 10.6.237.231
remote IP address 10.64.64.64
primary DNS address 202.96.128.86
secondary DNS address 202.96.134.133
sent [IPV6CP ConfReq id=0x1 <addr fe80::fc49:896b:7e79:5ad8>]
```

Figure 4.12-5 Successful IP acquisition

Use the ifconfig directive to see the ip address obtained.

ifconfig

```
root@atk-dlam62l:/opt/shell/4G# sent [IPV6CP ConfReq id=0x1 <addr fe80::fc49:896b:7e79:5ad8>]
ifconfig
eth0      Link encap:Ethernet  HWaddr 30:30:D0:97:26:CE
          UP BROADCAST MULTICAST  MTU:1500  Metric:1
          RX packets:0 errors:0 dropped:0 overruns:0 frame:0
          TX packets:0 errors:0 dropped:20 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)

eth1      Link encap:Ethernet  HWaddr DE:32:56:83:21:B4
          UP BROADCAST MULTICAST  MTU:1500  Metric:1
          RX packets:0 errors:0 dropped:0 overruns:0 frame:0
          TX packets:0 errors:0 dropped:19 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:65536  Metric:1
          RX packets:640 errors:0 dropped:0 overruns:0 frame:0
          TX packets:640 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:38400 (37.5 KiB)  TX bytes:38400 (37.5 KiB)

ppp0      Link encap:Point-to-Point Protocol
          inet addr:10.6.237.231  P-t-P:10.64.64.64  Mask:255.255.255.255
          UP POINTOPOINT RUNNING NOARP MULTICAST  MTU:1500  Metric:1
          RX packets:4 errors:0 dropped:0 overruns:0 frame:0
          TX packets:12 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:3
          RX bytes:52 (52.0 B)  TX bytes:170 (170.0 B)
```

Figure 4.12-6 EC20 PPP0 gets the IP address

Test your Internet connection by ping www.baidu.com. The -I argument specifies ppp0(4G network), and press Ctrl+c to end the ping. The result below indicates that you have Internet access.

ping www.baidu.com -I ppp0

```

root@atk-dlam62l:~# ping www.baidu.com -I ppp0
PING www.baidu.com (103.235.46.115): 56 data bytes
64 bytes from 103.235.46.115: seq=0 ttl=46 time=217.116 ms
64 bytes from 103.235.46.115: seq=1 ttl=46 time=238.072 ms
64 bytes from 103.235.46.115: seq=2 ttl=46 time=237.886 ms
64 bytes from 103.235.46.115: seq=3 ttl=46 time=237.796 ms
^C
--- www.baidu.com ping statistics ---
4 packets transmitted, 4 packets received, 0% packet loss
round-trip min/avg/max = 217.116/232.717/238.072 ms

```

Figure 4.12-7 PPP0 ping Baidu

If the error 'ping: bad address 'www.baidu.com'' occurs during the ping test, execute the following commands and try again.

```

echo "nameserver 8.8.8.8" > /etc/resolv.conf
echo "nameserver 8.8.4.4" >> /etc/resolv.conf

```

```

root@atk-dlam62l:/opt/shell/4G# ping www.baidu.com -I ppp0
ping: bad address 'www.baidu.com'
root@atk-dlam62l:/opt/shell/4G#

```

Figure 4.12-8 DNS missing error

4.12.2 quectel-CM Dial-up Internet Access

The ATK-DLAM62L development board has integrated the dial-up tools of Fibocom and Quectel. Directly execute the following commands for dial-up.

```
./opt/shell/4G/quectel-CM &
```

Note: You can use quectel-CM > /dev/null 2>&1 & to redirect logs to /dev/null, which means disabling logs.

```

root@atk-dlam62l:~# ./opt/shell/4G/quectel-CM &
root@atk-dlam62l:~# [01-01_08:04:13:943] QConnectManager_Linux_V1.6.8
[01-01_08:04:13:948] Find /sys/bus/usb/devices/1-1.4 idVendor=0x2c7c idProduct=0x125, bus=0x001, dev=0x004
[01-01_08:04:13:949] Auto find qmichannel = /dev/cdc-wdm0
[01-01_08:04:13:949] Auto find usbnet_adapter = wwan0
[01-01_08:04:13:949] netcard driver = qmi_wwan, driver version = 6.12.24
[01-01_08:04:13:949] Modem works in QMI mode
[01-01_08:04:13:957] cdc_wdm_fd = 7
[01-01_08:04:14:062] Get clientWDS = 5
[01-01_08:04:14:094] Get clientDMS = 1
[01-01_08:04:14:126] Get clientNAS = 2
[01-01_08:04:14:158] Get clientUIM = 1
[01-01_08:04:14:191] Get clientWDA = 1
[01-01_08:04:14:222] requestBaseBandVersion EC20CEF1LGR08A01M1G
[01-01_08:04:14:350] requestGetSIMStatus SIMStatus: SIM_READY
[01-01_08:04:14:413] requestGetProfile[pdp:1 index:1] CTNET///0/IPV4
[01-01_08:04:14:446] requestRegistrationState2 MCC: 460, MNC: 11, PS: Attached, DataCap: LTE
[01-01_08:04:14:479] requestQueryDataCall IPv4ConnectionStatus: DISCONNECTED
[01-01_08:04:14:479] ip addr flush dev wwan0
[01-01_08:04:14:486] ip link set dev wwan0 down
[01-01_08:04:14:541] requestSetupDataCall WdsConnectionIPv4Handle: 0x86d18890
[01-01_08:04:14:671] ip link set dev wwan0 up
[01-01_08:04:14:680] busybox udhcpc -f -n -q -t 5 -i wwan0
udhcpc: started, v1.36.1
udhcpc: broadcasting discover
udhcpc: broadcasting discover
udhcpc: broadcasting discover
udhcpc: broadcasting discover
udhcpc: no lease, failing
[01-01_08:04:29:926] File:ql_raw_ip_mode_check Line:143 udhcpc fail to get ip address, try next:
[01-01_08:04:29:926] ip link set dev wwan0 down
[01-01_08:04:29:935] echo Y > /sys/class/net/wwan0/qmi/raw_ip
[01-01_08:04:29:935] ip link set dev wwan0 up
[01-01_08:04:29:944] busybox udhcpc -f -n -q -t 5 -i wwan0
udhcpc: started, v1.36.1
udhcpc: broadcasting discover
udhcpc: broadcasting select for 10.18.206.205, server 10.18.206.206
udhcpc: lease of 10.18.206.205 obtained from 10.18.206.206, lease time 7200
[01-01_08:04:30:090] deleting routers
[01-01_08:04:30:119] adding dns 202.96.128.86
[01-01_08:04:30:119] adding dns 202.96.134.133

```

Figure 4.12-9 quectel-CM Dial-up

Use the ifconfig command to check the obtained IP address.

ifconfig

```

root@atk-dlam62l:~# ifconfig
eth0      Link encap:Ethernet  HWaddr 30:30:D0:97:26:CE
          UP BROADCAST MULTICAST  MTU:1500  Metric:1
          RX packets:0 errors:0 dropped:0 overruns:0 frame:0
          TX packets:0 errors:0 dropped:66 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)

eth1      Link encap:Ethernet  HWaddr 92:0A:A6:31:92:03
          UP BROADCAST MULTICAST  MTU:1500  Metric:1
          RX packets:0 errors:0 dropped:0 overruns:0 frame:0
          TX packets:0 errors:0 dropped:65 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:65536  Metric:1
          RX packets:816 errors:0 dropped:0 overruns:0 frame:0
          TX packets:816 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:48960 (47.8 KiB)  TX bytes:48960 (47.8 KiB)

wwan0     Link encap:UNSPEC  HWaddr 00-00-00-00-00-00-00-00-00-00-00-00-00-00-00-00
          inet addr:10.18.206.205  P-t-P:10.18.206.205  Mask:255.255.255.252
          inet6 addr: fe80::34d8:9f6d:a1e3:690a/64 Scope:Link
          UP POINTOPOINT RUNNING NOARP MULTICAST  MTU:1500  Metric:1
          RX packets:41 errors:0 dropped:0 overruns:0 frame:0
          TX packets:60 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:5795 (5.6 KiB)  TX bytes:6146 (6.0 KiB)

```

Figure 4.12-10 wwan0 interface

To verify Internet connectivity, you can ping an external network. Run the command to ping Baidu. The -I option is used to specify the network interface.

ping www.baidu.com -I wwan0

```

root@atk-dlam62l:~# ping www.baidu.com -I wwan0
PING www.baidu.com (183.2.172.17): 56 data bytes
64 bytes from 183.2.172.17: seq=0 ttl=52 time=30.226 ms
64 bytes from 183.2.172.17: seq=1 ttl=52 time=50.062 ms
64 bytes from 183.2.172.17: seq=2 ttl=52 time=49.929 ms
64 bytes from 183.2.172.17: seq=3 ttl=52 time=49.802 ms
^C
--- www.baidu.com ping statistics ---
4 packets transmitted, 4 packets received, 0% packet loss
round-trip min/avg/max = 30.226/45.004/50.062 ms
root@atk-dlam62l:~#

```

Figure 4.12-11 Ping Baidu for testing

4.12.3 GPS test

EC20 also has a GPS function, which can be set and turned on through the AT command. This section only teaches how to turn on and view the original GPS data, and other AT commands move company information.

Note: There is a GPS antenna interface in Figure 4.12.1 EC20 installation. EC20 with GPS antenna can be purchased. The antenna needs to be connected to the GPS interface.

The 4G module is normally initialized with the following printing information:


```

[ 5.488336] usbcore: registered new interface driver option
[ 5.494420] usbserial: USB Serial support registered for GSM modem (1-port)
[ 5.501815] option 1-1.4:1.0: GSM modem (1-port) converter detected
[ 5.508811] usb 1-1.4: GSM modem (1-port) converter now attached to ttyUSB0
[ 5.516169] option 1-1.4:1.1: GSM modem (1-port) converter detected
[ 5.523118] usb 1-1.4: GSM modem (1-port) converter now attached to ttyUSB1
[ 5.530432] option 1-1.4:1.2: GSM modem (1-port) converter detected
[ 5.537354] usb 1-1.4: GSM modem (1-port) converter now attached to ttyUSB2
[ 5.544689] option 1-1.4:1.3: GSM modem (1-port) converter detected
[ 5.552290] rtk_btusb: btusb_probe intf->cur_altsetting->desc.bInterfaceNumber 0
[ 5.552621] usb 1-1.4: GSM modem (1-port) converter now attached to ttyUSB3
[ 5.570092] usbcore: registered new interface driver rtk_btusb
[ 5.580859] rtk_btusb: btusb_open hdev->promisc ==0
[ 5.585872] rtk_btusb: chip type value: 0x76

```

Figure 4.12-12 The 4G module is initialized normally

You can see above that you have initialized 4 serial ports. Run the following command to use GPS function.

```

echo -e "AT+QGPS=1\r\n" >/dev/ttyUSB2
// Send the AT+QGPS=1 data to USB2 by echo command
cat /dev/ttyUSB1

```

```

root@atk-dlam62l:~# echo -e "AT+QGPS=1\r\n" >/dev/ttyUSB2
root@atk-dlam62l:~# cat /dev/ttyUSB1
$GPVTG,,T,,M,,N,,K,N*2C
$GPGSA,A,1,,,,,,,,,,,,,*32
$GPGGA,,,,,0,,,,,,,,*66
$GPRMC,,V,,,,,,,,,N*53
$GPVTG,,T,,M,,N,,K,N*2C
$GPGSA,A,1,,,,,,,,,,,,,*32
$GPGGA,,,,,0,,,,,,,,*66
$GPRMC,,V,,,,,,,,,N*53
$GPVTG,,T,,M,,N,,K,N*2C

```

Figure 4.12-13 GPS raw data

Disabling GPS allows you to run the following code: You can terminate the cat command by pressing Ctrl + c.

```

echo -e "AT+QGPSEND\r\n" >/dev/ttyUSB2
cat /dev/ttyUSB1

```

At this time we can see that ttyUSB1 serial port no longer prints the original GPS data.

4.13 4G module ME3630-W test

4G module interface is reserved on ATK-DLAM62L, ME3630-W, EC20 and other 4G modules are installed. Insert the ME3630-W module into the interface of the 4G module and screw it on. Ensure that the 4G module is connected to the seat interface. Please use the original antenna and connect the antenna to the MAIN interface of the 4G module. (Warm tip: after the 4G module is unsealed, the gold finger is easy to oxidize. After a long time, there will be a layer of oxide film on it. If many experiments are not successful, please check whether the 4G module is inserted or whether there is an oxide film. Please save the 4G module after use!)

Insert the 4G card correctly (support mobile, China Unicom, Telecom 4G card) and insert the module, and the WWAN LED on the bottom board will light green after the development board is started. If the green light of the WWAN LED does not light up, please check whether the module is properly connected and inserted, and whether the 4G card is inserted.

Insert the 4G SIM card correctly (supports China Mobile, China Unicom, and China Telecom 4G cards) and properly install the module. After the development board boots up, the WWAN LED on the baseboard will light up green.

If the WWAN green LED does not turn on, please check whether the module is properly connected and inserted, and whether the 4G SIM card is inserted correctly.

After the 4G module is loaded normally, three ttyUSB* devices will be generated under /dev/.

The 4G module communicates via AT commands through /dev/ttyUSB2.

```
ls /dev/ttyUSB*
```

4.13.1 pppd dial-up Internet access

The factory file system /opt/shell/4G directory has been stored in the test 4G module related scripts. Create a new directory that will be used to generate DNS client configuration files when the following script is run.

```
mkdir /etc/ppp/          # If it already exists, you don't need to duplicate it
```

Go to /opt/shell/4G.

```
cd /opt/shell/4G
```

Use the ls command to see that there are multiple script files in this path. If there are no such files, please update your system in time.

Execute the following ppp-on script with & running in the background.

```
./ppp-on-10086 &
```

You can use ifconfig to check the network name of the 4G module.

```
ifconfig
```

Ping Baidu to test connectivity. The -I option is used to specify the network interface.

If you see the result below with data replies, it means the connection is successful. Press "Ctrl + C" to terminate the command.

```
ping www.baidu.com -I ppp0
```

If the error ping: bad address 'www.baidu.com' occurs during the ping test, execute the following commands and try again:

```
echo "nameserver 8.8.8.8" > /etc/resolv.conf
echo "nameserver 8.8.4.4" >> /etc/resolv.conf
```

4.13.2 ECM access to the Internet

From the perspective of the operating system, a CDC ECM device acts as a virtual Ethernet network adapter, which includes the MAC address and IP address required by a standard network interface card.

ECM_DEMO_AUTO and ECM_DEMO are high emerging ECM online programs (we don't need to know the source code, because it's just a tool). The tool has been placed in /opt/shell/4G. Please enter the /opt/shell/4G directory.

```
cd /opt/shell/4G
```


The instructions are as follows:

```
./ECM_DEMO -t up           // Enable ecm Internet access
./ECM_DEMO -t down        // Turn off the ecm connection
./ECM_DEMO -t up -p /dev/ttyUSB1 -a 3gnet
// When enabled, specify the appropriate apn and dial-in port
./ECM_DEMO -t down -p /dev/ttyUSB1 // Turn off the ecm connection
```

The parameters can be viewed in the source code or by running `ECM_DEMO-h`

`ECM_DEMO_AUTO` and `ECM_DEMO` parameters are the same, the difference is that `ECM_DEMO` executes the process will exit, `ECM_DEMO_AUTO` will run and every once in a while will check whether the network is disconnected, the network will automatically reconnect

`ECM_DEMO_AUTO` executes the dialing process automatically by default, so it is suitable for booting programs.

To configure the ECM connection, if you have `pppd` running, run the `disconnect` script to disconnect the `pppd` dial-up connection, and then run the following command to configure the ECM connection.

```
./disconnect
./ECM_DEMO -t up
```

Use the `ifconfig` command to check the ip address obtained. If the ip address is not obtained, use `udhcpc -i usb0` to obtain it.

```
ifconfig
```

We can instruct the `usb0` node to ping Baidu.

```
ping www.baidu.com -I usb0
```

4.14 TF(SD) card test

Instruction Prompt:

The `time` command is often used to measure the running time of a command, `dd` is used to copy, read from the `if`(input file) file and write to the `of`(output file) specified file, `bs` is the block size of each write, and `count` is the number of read and write blocks. `if=/dev/zero` does not produce IO, that is, it can continuously output data, so it can be used to test pure write speed.

After the development board is started, insert the TF card into the slot at the bottom of the development board. Development board will automatically mount TF card. (Note that you need to use FAT32 format TF card, not NTFS format). Use the `df -h` command to determine the partition on which SD is mounted. Here is the result:

If the inserted TF card cannot be mounted automatically, you can manually execute the `mount` command:

```
mount /dev/mmcblk1p1 /mnt/sdcard
df -h
```

```

root@atk-dlam62l:~# mount /dev/mmcblk1p1 /mnt/sdcard
[ 54.843553] FAT-fs (mmcblk1p1): Volume was not properly unmounted. Some data may be corrupt. Please run fsck.
root@atk-dlam62l:~# df -h
Filesystem      Size      Used Available Use% Mounted on
/dev/root        7.0G      586.6M      6.0G   9% /
devtmpfs         218.0M          0      218.0M   0% /dev
tmpfs            235.5M          0      235.5M   0% /dev/shm
tmpfs            235.5M      112.0K      235.4M   0% /tmp
tmpfs            235.5M      184.0K      235.3M   0% /run
/dev/mmcblk0p1   126.0M       58.1M       68.0M  46% /boot
/dev/mmcblk1p1   29.5G       50.1M      29.5G   0% /mnt/sdcard
root@atk-dlam62l:~#

```

挂载出来的sd卡分区

Figure 4.14-1 df-h to view the partition

Read speed test

Often used to test the hard disk read speed instructions have hdparm and dd, this only demonstrates hdparm read TF card speed. (The reading speed depends on the type of TF card you are using)

```
hdparm -t /dev/mmcblk1p1
```

```

root@atk-dlam62l:~# hdparm -t /dev/mmcblk1p1
/dev/mmcblk1p1:
Timing buffered disk reads: 72 MB in 3.02 seconds = 24336 kB/s
root@atk-dlam62l:~#

```

Figure 4.14-2 Read test

It can be seen that the read speed is 30301KB/s.

Write speed test

```
time dd if=/dev/zero of=/mnt/sdcard/test bs=1024k count=100 && sync
```

```

root@atk-dlam62l:~# time dd if=/dev/zero of=/mnt/sdcard/test bs=1024k count=100 && sync
100+0 records in
100+0 records out
real    0m 2.83s
user    0m 0.00s
sys     0m 0.69s
root@atk-dlam62l:~#

```

Figure 4.14-3 Writing tests

Delete the test file in the directory "/mnt/sdcard".

4.15 ADC testing

Note: The maximum absolute value of ADC acquisition voltage is 1.8V, please do not exceed 1.8V, otherwise it may cause damage to the chip. ATK-DLAM62L has an ADC at RP1 to facilitate user testing. The potentiometer is connected to ADC0_AIN1 interface, as shown below:

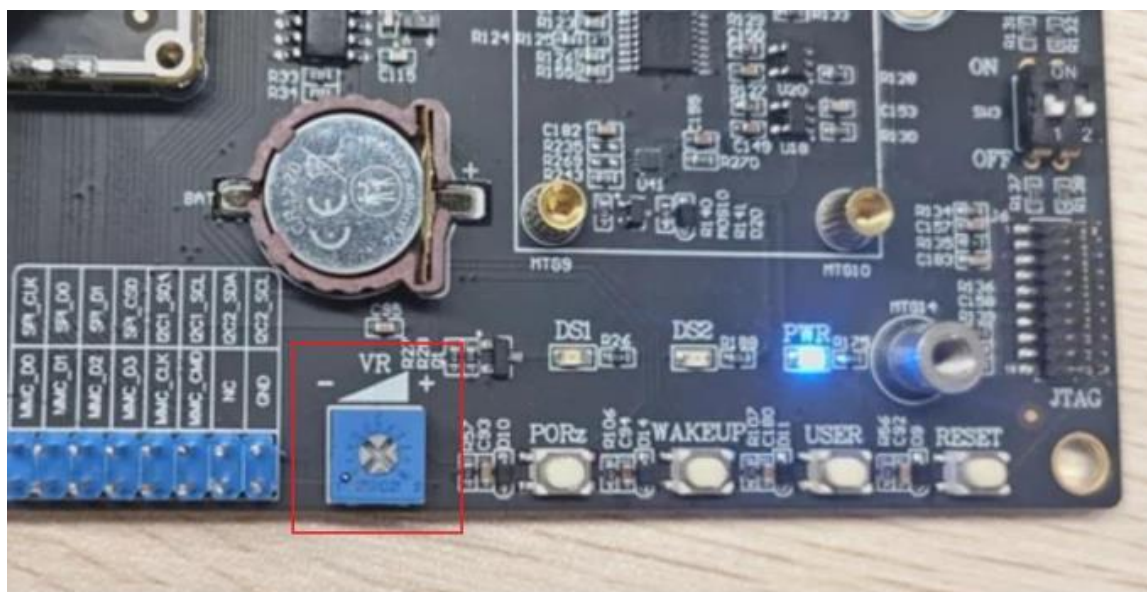


Figure 4.15-1 Potentiometer diagram

The following commands can be executed in the factory system to view the raw ADC values:

```
cat /sys/bus/iio/devices/iio\:device0/in_voltage1_raw
```

```
root@atk-dlam62l:~# cat /sys/bus/iio/devices/iio\:device0/in_voltage1_raw
4063
root@atk-dlam62l:~# cat /sys/bus/iio/devices/iio\:device0/in_voltage1_raw
4054
root@atk-dlam62l:~# cat /sys/bus/iio/devices/iio\:device0/in_voltage1_raw
4059
root@atk-dlam62l:~# cat /sys/bus/iio/devices/iio\:device0/in_voltage1_raw
4074
```

Figure 4.15-2 Check the ADC voltage value

The formula for calculating the ADC voltage is:

$$D = V_{in} * (2^n - 1) / V_{ref}$$

Explanation:

D = Raw ADC value

V_{in} = Input voltage value

n = Resolution, the ADC resolution of AM62L is 12

V_{ref} = Reference voltage value, which is 1.8V

4.16 FAN Interface Test

A cooling fan is required for testing the FAN interface. Alternatively, an oscilloscope can be used to measure the 4th pin of the FAN interface to observe the PWM waveform.

Connect the cooling fan module to the FAN interface of the development board. The connection with the development board is shown in the figure below:



Figure 4.16-1 Connect the development board.

The following commands can be executed to test the cooling fan. There are 7 gears in total, where 0 is the lowest gear and 7 is the highest gear.

```
echo 0 > /sys/class/thermal/cooling_device0/cur_state
echo 7 > /sys/class/thermal/cooling_device0/cur_state
```

4.17 CPU Temperature

Use the following command to check the CPU temperature. The temperature level is related to the ambient temperature and the operating status of the CPU. For example, when running large applications, videos, or audio, the temperature will be relatively higher.

It is not recommended to touch the surface of the CPU with your hands to prevent chip damage caused by static electricity.

```
cat /sys/class/thermal/thermal_zone0/temp
```

The running result is shown in the figure below:

```
root@atk-dlam62l:~# cat /sys/class/thermal/thermal_zone0/temp
50602
root@atk-dlam62l:~#
```

Figure 4.17-1 Check CPU Temperature

4.18 CPU Frequency

CPU frequency cannot be viewed or changed temporarily.

4.19 QT Interface

The ATK-DLAM62L development board runs the QT interface by default.

4.19.1 Temporarily Close QT Interface

To temporarily close the QT interface, run the following commands:

```
/etc/init.d/S99weston stop  
fbinit
```

The first command is to temporarily close QT. The second command clears the display interface of the screen. Run the above code and you will see the screen turn black.

4.19.2 QT Permanent Disable Manual

To permanently disable QT, simply move the files **S98weston** and **S99systemui** out of the `/etc/init.d/` directory.

4.20 Serial Port Issues

When entering long commands, you may find that the text does not wrap to a new line. To solve this problem, run the following code after the terminal starts:

```
resize  
export TERM=xterm
```

After input is complete, the terminal will be able to accept long commands.